



ECG-RSI: Recursive Self-Improvement through Targeted ECG Waveform Evolution

Wang Xiaoliang^{♥♣*}, Zhuo Chen^{♣*}, Chuanyang Zheng[♣], Jiangwei Lao[♣], Quanbin Wang[♣]

Honglin Lin^{♠♦}, Jiajia Liu[♣], Zhihong Lu[♣], Hao Jin[♥], Jian Wang[♣], Lijun Wu^{♠♦†}

♥ Zhejiang University ♣ Ant Group ♠ Shanghai Artificial Intelligence Laboratory ♦ Shanghai Jiao Tong University

 github.com/pork-pay/ECG-RSI | [Project page](#)

Abstract

Electrocardiogram (ECG) models need supervision that targets evolving weaknesses and remains faithful to waveform evidence. We introduce **ECG-RSI**, a recursive self-improvement framework combining verified waveform synthesis with feedback-guided intervention search. Signal remeasurement verifies waveform edits; construction errors guide generator repair, while learner errors motivate interventions compared on a separate selection set. Across seven benchmarks, three rounds improve Qwen3.5-35B-A3B by **21.47 percentage points in mean QA accuracy** and **9.77 points in mean classification F1**. Selected interventions shift from direct reinforcement to confusion-focused contrasts, suggesting that **updates can change which supervision is most valuable next**. Continued feedback correction yields diminishing external gains. RSI should therefore **separate error diagnosis from intervention selection**, renewing how it teaches as the learner evolves and assessing progress through external transfer.

1 Introduction

Electrocardiogram (ECG) interpretation requires connecting **waveform patterns** to **measurable features** and clinically meaningful conclusions. Multimodal models have begun to learn this connection from ECG images and instruction data [1, 2, 3]. However, improving ECG understanding requires more than increasing the number of training questions: each answer must be supported by the waveform presented to the model. Changes in heart rate, intervals, or morphology can alter the interpretation of related ECG findings.

Existing studies improve ECG supervision through instruction construction and evidence-aware learning [1, 2, 3], while conditional signal generators expand waveform diversity [4, 5]. Yet it remains difficult to turn a learner’s specific weakness into a controlled training example. A generated waveform must express the intended ECG feature without introducing conflicting changes, and its question and answer must agree with the resulting signal. Moreover, the examples most useful to a model can change after each training round: a fixed data mixture cannot respond to newly identified weaknesses, persistent confusions, or shifts in task difficulty. Our central insight is to use **learner feedback** to specify ECG evidence and **signal measurements** to verify it.

We propose **ECG-RSI**, a framework that connects targeted ECG data synthesis with feedback-driven recursive self-improvement (Figure 1). Its data factory generates questions from existing signals and uses constrained reverse perturbations to synthesize waveforms with requested feature contrasts. The resulting waveforms are remeasured before their questions and answers are accepted. Construction failures guide revisions to perturbation operators

* Equal contribution. † Corresponding author.

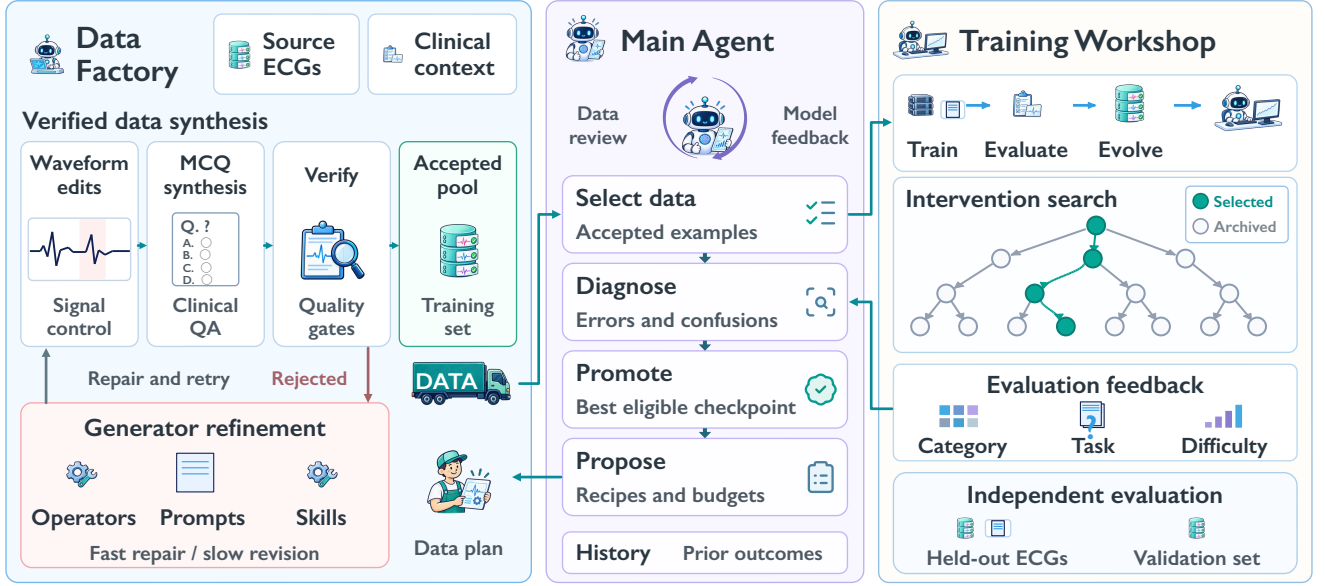


Figure 1 ECG-RSI: feedback-driven data and model evolution. The data factory constructs and verifies ECG questions, with rejected candidates driving local revisions of operators, prompts, and skills. The main agent diagnoses feedback, proposes interventions, and selects accepted data for training. The training workshop evaluates candidate checkpoints and supplies category-, task-, and difficulty-level feedback. Independent evaluation is separated from the update loop. The search tree and highlighted path illustrate candidate comparison and model selection.

and question-generation workflows, while the learner’s errors guide the categories, tasks, and difficulty levels of subsequent training data. At each round, candidate data recipes are used to train and compare model branches, and the selected branch informs the next round of synthesis and training.

Our contributions are threefold:

- We propose **ECG-RSI**, a framework that connects ECG data synthesis with recursive model improvement, enabling training data to adapt to the learner’s changing capability gaps.
- We develop **target-driven reverse waveform synthesis**, which translates desired ECG feature contrasts into constrained signal perturbations and verifies the resulting waveforms to provide measurement-grounded supervision.
- We design a **dual-feedback evolution loop** in which construction feedback improves the data-generation process, while learner feedback guides subsequent data selection and model updates.

2 Related Work

2.1 ECG Multimodal Learning and Evidence-Grounded Supervision

Medical vision-language foundation models provide a broader basis for domain adaptation. MedGemma adapts Gemma 3 models for **medical image and text understanding** and supports further task-specific fine-tuning [6].

ECG-language models have progressed from **ECG instruction tuning** and multimodal alignment toward more flexible question answering and **clinically grounded interpretation**. PULSE, a multimodal large language model for ECG image comprehension, is instruction-tuned on ECGInstruct, an ECG image instruction dataset, and evaluated on ECGBench, a benchmark for ECG image interpretation [1]. GEM links signal and image representations to measurable ECG evidence [2]. ECG-R1 incorporates clinical protocols and diagnostic-evidence supervision into ECG interpretation [3]. Subsequent work broadens the types of questions, input formats, and supervision available to ECG models: anyECG-chat supports flexible ECG inputs and multiple tasks [7], Q-Heart introduces knowledge-informed ECG question answering [8], ECG-LLM uses clinically structured questions for cardiac reasoning [9], and ELF studies a simpler encoder-free ECG-language architecture [10]. MEETI aligns signals, images, extracted features, and interpretations within a multimodal ECG dataset [11], while ECG-Expert-QA provides diagnostically diverse questions for evaluation [12]. ECG input representations also affect performance [13].

Improved task performance does not, however, establish that a model has interpreted the ECG evidence correctly. ECG Mirage probes whether predictions depend on the **patient-specific ECG** [14]. ECG-Reasoning-Benchmark examines stepwise ECG interpretation [15], and recent work evaluates ECG reasoning by separating waveform perception from clinical deduction [16]. Image-based model evaluations likewise expose the difficulty of reliable ECG interpretation from rendered tracings [17]. These studies motivate supervision that remains tied to the waveform presented to the learner. ECG-RSI addresses this need by constructing requested feature contrasts and deriving questions and answers from measurements of the resulting signals.

2.2 Conditional ECG Synthesis and Controlled Interventions

ECG synthesis has advanced through **disease-conditioned** adversarial generation [4], conditional diffusion [18, 19], clinical-text conditioning [20], and report- and patient-conditioned generation [5]. MIDT-ECG and FlowECG further investigate **waveform fidelity**, inter-lead coherence, and generation efficiency [21, 22]. More directly related to controllability, ECGTwin generates condition-guided personalized ECGs while seeking to preserve individual characteristics [23], and TransDiffECG explores semantically controllable signal generation [24]. Generative counterfactual ECGs have also been used to study how waveform changes affect an AI-ECG model’s predictions [25]. These works establish several ways to generate or alter plausible ECGs, but they address different objectives from constructing a training example for a **specified learner weakness**. ECG-RSI starts with a training-side waveform and a requested feature contrast, applies a constrained intervention, and remeasures the result. It accepts an example only when the intended change is achieved and relevant non-target properties are preserved; its question and answer are then grounded in the **achieved waveform**. This grounds supervision in controlled signal changes.

2.3 Feedback-Driven Data Evolution and Recursive Self-Improvement

Earlier self-improvement methods revise responses or retain verbal feedback [26, 27], generate instruction data [28, 29], or iteratively refresh reasoning supervision [30]. More recent systems connect generated tasks to **learner improvement**: CoEvolve guides task synthesis using learner behavior [31], NeoHorse-1 adapts post-training mixtures using capability evidence [32], and Absolute Zero and VisPlay couple task generation with solver improvement [33, 34]. RSI-Bench-Data places data construction and selection at the center of RSI evaluation [35]. Related studies examine whether a self-synthetic pipeline continues to provide learnable information [36], while VISA revises multimodal instruction synthesis using verification and **target-model failures** [37]. Recursive on-policy distillation co-evolves the privileged teacher with the student [38], while RLTL;DR internalizes concise feedback from failed attempts [39]. Both make the supervision produced by an earlier attempt useful to subsequent learning. A second line evolves the mechanism that performs improvement. STOP and the Darwin Gödel Machine revise improver code or agent workflows [40, 41]; MetaSkill-Evolve updates skills and their management mechanisms [42]; and Dream-RSI and EvoX study evolving environments or search strategies [43, 44]. DataFoundry evolves data preparators [45], Astar proposes subsequent evolution directions [46], and DataEvolver uses rejected samples to revise a data-construction process [47]. Hyperagents and RRSI investigate agent-level improvement mechanisms and the selection of their revisions [48, 49]. Learning to Self-Evolve studies how feedback can guide subsequent adaptation [50], while EvoOntology and MediSkill-Evo illustrate evolution of structured agent knowledge and clinically constrained process knowledge [51, 52]. RecurSE additionally studies evolving self-evaluation [53]. AIDE² retains agent-code revisions using hidden evaluations, while SIFT uses pairwise candidate rankings to focus costly evaluations [54, 55]. ScienceBuddy alternates harness evolution with model learning [56]. In medicine, MedZERO couples task generation with evidence-grounded reasoning and controlled knowledge accumulation [57].

These approaches make candidate evaluation and knowledge retention central to continued improvement. Recent work also studies when to stop evolution, using paired evaluation outcomes to detect diminishing returns and select an earlier artifact [58]. ECG-RSI examines intervention selection through measurable ECG evidence: construction feedback repairs synthesis, learner feedback proposes training directions, and separate selection and external evaluation assess their utility.

3 Method

ECG-RSI couples **verified synthesis** with **feedback-driven training** (Figure 1). Construction feedback improves waveform editing and question generation; learner feedback selects data for subsequent model updates. The loop

separates example validity from learner-specific utility [35].

3.1 Framework and Feedback Interfaces

System state and data boundaries. At round t , the system state comprises the accepted learner θ_t , retained training pool D_t , and **construction policy** ϕ_t . The policy contains retrieval constraints, waveform operators, prompts, and reusable rules, separate from learner weights [40]. Source ECGs and morphology donors are restricted to training data and are **patient-disjoint** from development and independent evaluation data.

V_{fb} diagnoses learner errors, while a separate V_{sel} governs **model selection**. Independent evaluation data guide neither generation nor selection.

Two feedback loops. **Construction feedback** identifies invalid waveform edits or questions and guides generator repair. **Learner feedback** guides recipes using gaps in ECG capability, question type, difficulty, and confusion pairs. For a selected update path, one round can be summarized as

$$\begin{aligned} e_t &= \text{Diagnose}(\theta_t, V_{fb}), \\ q_t &= \text{Plan}(e_t, D_t; B_t), \\ (A_t, L_t, \phi_{t+1}) &= \text{ConstructValidate}(q_t, S_{\text{train}}, \phi_t), \\ \tilde{\theta}_{t+1} &= \text{Train}(\theta_t, A_t \cup R_t), \\ \theta_{t+1} &= \text{Promote}(\theta_t, \tilde{\theta}_{t+1}; V_{sel}). \end{aligned} \tag{1}$$

Here B_t is the construction budget, A_t the accepted new examples, L_t the construction log, and R_t the replay data. Equation (1) describes one candidate; sibling recipes use the **same parent** with separate examples and policy versions. Promotion adds the winning examples to D_t ; otherwise, the parent and retained pool remain unchanged.

3.2 Targeted ECG Synthesis with Measurement-Grounded Supervision

3.2.1 Targeted Waveform Construction

Forward synthesis derives questions, answers, and explanations from measured training ECGs. **Reverse synthesis** edits an eligible waveform to express a requested feature contrast, then remeasures the signal before constructing supervision. Both routes check evidence, context, answer uniqueness, and explanation support.

Reverse requests come from learner weaknesses or distractors in training questions. A normal-versus-prolonged PR contrast requires a measurable target and waveform edit, not relabeling.

Target selection and constrained editing. A delineator extracts $u = f(S)$ from ECG S : heart rate (HR), R–R interval (RR), P-onset-to-QRS-onset interval (PR), QRS duration, QRS-onset-to-T-end interval (QT), corrected QT (QTc), and amplitudes. Feature k on source i receives score $s_{ik} = a_{ik}r_kd_k$, combining **definability** a_{ik} , extraction reliability r_k , and diagnostic utility d_k . Features above the registered threshold are eligible. Appendix A.6 gives measurement conversions and the weighting formula, including the outstanding scale-normalization specification.

An eligible request specifies the **target range**, protected features and tolerances, permitted operators and donors, and attempt budget. The policy applies an operator sequence $S' = T_\delta(S)$ after checking applicability. PR edits require P and QRS onsets; complex diagnoses require morphology beyond interval changes (Figure 2).

3.2.2 Validation and Generator Repair

After editing, the system re-extracts features $u' = f(S')$. Let k be the target feature and Δ_k its requested change. We record the **target error**

$$E_{\text{target}} = \left| [f_k(S') - f_k(S)] - \Delta_k \right|$$

separately from the drift of each protected feature j ,

$$E_{\text{protect},j} = |f_j(S') - f_j(S)|.$$

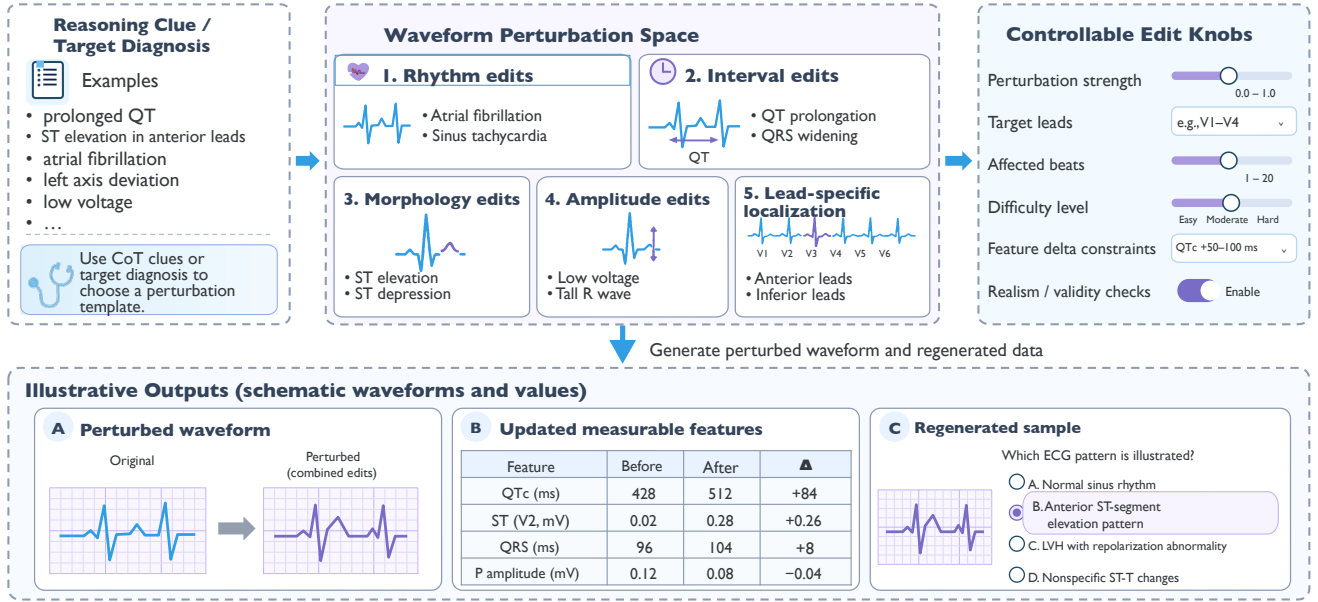


Figure 2 Controllable waveform perturbations across rhythm, intervals, morphology, amplitude, and individual leads. The examples illustrate how an intended feature change is translated into a localized signal edit. Edited waveforms are remeasured to check the target and protected features before question generation, linking controllable signal changes to evidence-grounded supervision.

Acceptance requires target and protected-feature errors to satisfy their **respective tolerances**, with checks of boundary continuity, unedited regions, lead consistency, morphology, leakage, diversity, and cost.

The accepted signal is rendered for question, answer, context, and explanation generation. Review requires a **unique supported answer** and evidence for each claimed observation, so supervision follows the remeasured signal.

Construction failure and generator repair. Failures guide component repair, as in data-preparator revision [45]. **Signal failures** trigger operator repair; missing morphology triggers applicability or retrieval changes; ambiguity triggers question/context revisions; unsupported explanations trigger prompt repair.

Repairs record their trigger, changed rule, validation result, and scope. Old and revised versions are compared on **matched sources, requests, seeds, and budgets**; adoption requires signal and question checks to pass. Appendices A.2 and A.3 describe the rules and recording protocol; Appendix D provides stage prompts.

3.3 Feedback-Driven Recursive Model Improvement

3.3.1 Learner Diagnosis and Data Recipes

Feedback errors, invalid outputs, and regressions guide **recipes** over ECG category g , learner-relative difficulty d , and supervision representation s :

$$\pi_t^a(g, d, s) = p_t^a(g) p_t^a(d | g) p_t^a(s | g, d). \quad (2)$$

Difficulty is **learner-relative** [33]. For training-side question i , K_i independent answers under fixed decoding settings give $\hat{p}_{it} = K_i^{-1} \sum_{r=1}^{K_i} \mathbf{1}[\hat{y}_{ir} = y_i]$, used to group reliably solved, mixed-success, and unresolved examples. Probe budgets, scoring, output-validity requirements, and thresholds are fixed before branch comparison.

Supervision can retain the teacher explanation, expand intermediate ECG evidence, or add checks targeting an observed learner error. As in iterative reasoning supervision [30], explanations supply training targets; paired variants must preserve the **verified image, question, options, answer, and clinical facts**.

Recipes also fix source/edit constraints, replay, and training budgets. Recipe selection changes the data mixture; operator or prompt repair changes ϕ_t . These interventions are tracked separately (Appendix A.1).

3.3.2 Same-Parent Branch Exploration

Feedback errors and training-side probes instantiate three recipe families from the **same parent** θ_t , following empirical branch exploration [41, 43]:

- **A: Direct reinforcement.** Increase quotas for weak categories and question types while retaining related-category controls: an AV-block recipe emphasizes conduction measurements and diagnostic interpretation.
- **B: Confusion contrast.** Sample recurrently confused category pairs; construct options and explanations around their distinguishing ECG evidence. Regional STEMI contrasts, for example, emphasize lead-specific localization.
- **C: Difficulty and task curriculum.** Reweight difficulty bins and question types using learner probes, e.g., increasing L3, multistep, or evidence-chain questions. Curriculum denotes a branch-specific mixture.

Recipes specify **sampling quotas and construction requests**: retrieve eligible waveforms or apply permitted edits, then run the shared measurement and question-validity gates (Section 3.2.2). Category, difficulty, and task constraints can overlap; Table 13 lists each round’s actual mixtures.

Each branch retains the same historical training manifest and adds **10,000 previously unused accepted questions**, without an extra shared baseline set. All branches use three epochs of supervised fine-tuning (SFT) with matched settings; the language model is updated while the vision encoder and multimodal aligner remain frozen. For ECG image I , question Q , optional context C , and explanation–answer target y , the **training loss** is

$$\mathcal{L}_{\text{MCQ}} = - \sum_{\ell} \log p_{\theta}(y_{\ell} | I, Q, C, y_{<\ell}). \quad (3)$$

Replay and new-data exposure are recorded in examples and tokens; training settings appear in Appendix A.5.

3.3.3 Model Selection and Retained Lineage

After matched SFT, evaluate candidates on V_{sel} . Apply the predesignated **output-validity and non-target regression tolerances**, then retain the highest-scoring qualifying checkpoint and its data [49]. Archive other branches; if none qualifies, retain the parent.

Reevaluate the retained learner on V_{fb} to construct new recipes: **directions are reselected each round**. Figure 3 illustrates this process; Section 5.7 reports the observed choices.

4 Experiments

We evaluate benchmark performance, round-wise gains, and the training value of feedback-guided synthesis. The analysis asks how feedback granularity affects data utility, which candidate directions yield the strongest updates, and whether reductions in **feedback errors** transfer to external benchmarks.

Each benchmark retains its accuracy or F1 score (the harmonic mean of precision and recall); the seven-benchmark aggregate score (S7) is the **equal-weight mean** of four accuracy and three F1 scores on a 0–100 scale.

4.1 Main Benchmark Results

Table 1 compares the recorded scores of closed-source, open-source, and ECG-specific models with ECG-RSI on seven benchmarks. ECG-RSI R3 has the **highest tabulated S7 (46.61)**, with task-specific leaders.

4.2 Evolution across Rounds

Table 2 and Figure 4 show the R0–R3 trajectory of series R-A. Under the current protocol, the winning branch adds 10,000 previously unused accepted questions to the **cumulative training set** each round, alongside replay of all previous training data. Appendix B provides feedback breakdowns.

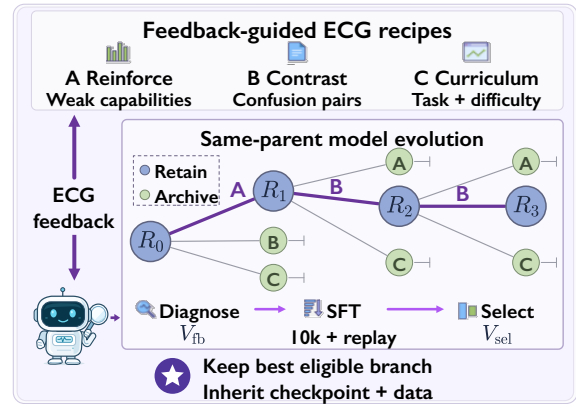


Figure 3 Schematic of feedback-guided model evolution. A/B/C change category coverage, confusion contrasts, or difficulty/task mixtures. Each candidate adds 10,000 accepted questions plus replay. Purple shows an illustrative lineage. Table 13 reports measured mixtures, scores, and choices.

Table 1 Comparison of ECG-RSI with proprietary, open-source, and ECG-specific models on seven benchmarks. The first four columns report question-answering accuracy and the next three report classification F1; S7 is their equal-weight mean on a 0–100 scale. ECG-RSI R3 achieves the highest aggregate score, while the strongest model varies by individual task.

Model / method	MMMU Acc.	CSN Acc.	G12 Acc.	ECGQA Acc.	PTB F1	CPSC F1	CODE15 F1	S7
<i>Proprietary models</i>								
GPT-5.6-sol	<u>56.00</u>	58.70	60.20	<u>32.10</u>	37.40	24.30	42.80	44.50
Gemini-3.8-Flash	59.10	<u>57.90</u>	<u>59.60</u>	33.40	<u>36.80</u>	<u>23.70</u>	<u>40.90</u>	<u>44.49</u>
<i>Open-source and base models</i>								
✦ LINGSHU-7B [59]	<u>47.60</u>	<u>45.20</u>	46.80	26.30	<u>33.87</u>	<u>18.60</u>	<u>36.50</u>	<u>36.41</u>
✦ LINGSHU-32B [59]	51.30	48.90	50.40	28.70	39.45	20.80	38.20	39.68
✦ HuatuoGPT-V-7B [60]	44.80	41.60	43.20	25.40	31.15	17.30	33.90	33.91
Qwen3.5-35B-A3B	42.00	30.91	41.36	23.99	29.50	13.10	30.20	30.15
✦ ECG-R1 [3]	38.12	44.87	<u>47.05</u>	<u>28.63</u>	22.91	17.88	34.55	33.43
✦ PULSE [1]	39.24	45.16	<u>43.87</u>	21.55	26.08	14.92	26.82	31.09
<i>ECG-RSI</i>								
✦ ECG-RSI R0	<u>42.00</u>	<u>30.91</u>	<u>41.36</u>	<u>23.99</u>	<u>29.50</u>	<u>13.10</u>	<u>30.20</u>	<u>30.15</u>
✦ ECG-RSI R3	57.50	67.23	66.24	33.18	38.30	22.80	41.00	46.61

All scores are on a 0–100 scale. Bold and underlining mark the best and second-best values within each model group; ties share a rank. Blue shading highlights our final model. ✦ denotes medical specialization. Qwen3.5-35B-A3B is the initial model (ECG-RSI R0). ECG-R1 and PULSE are ECG-specific comparison models.

Table 2 Round-wise benchmark performance of series R-A. R0 is the starting model; each accepted update adds 10,000 new questions while retaining historical training data. Bold and underlining mark the best and second-best values within each column; blue/red arrows indicate increases/decreases from the previous round, and blue shading marks R3. Aggregate performance improves through R3, although several individual metrics peak at R2.

R-A round	MMMU Acc.	CSN Acc.	G12 Acc.	ECGQA Acc.	PTB F1	CPSC F1	CODE15 F1	S7
R0	42.00	30.91	41.36	23.99	29.50	13.10	30.20	30.15
R1	48.00↑	61.68↑	64.61↑	<u>33.56</u> ↑	32.90↑	<u>23.00</u> ↑	38.80↑	43.22↑
R2	<u>56.00</u> ↑	<u>64.86</u> ↑	66.39 ↑	34.55 ↑	<u>36.40</u> ↑	23.90 ↑	41.50 ↑	<u>46.23</u> ↑
R3	57.50 ↑	67.23 ↑	<u>66.24</u> ↓	33.18↓	38.30 ↑	22.80↓	<u>41.00</u> ↓	46.61 ↑
Δ(R3–R0)	+15.50↑	+36.32↑	+24.88↑	+9.19↑	+8.80↑	+9.70↑	+10.80↑	+16.46↑

R0 is the starting model in R-A. Δ(R3–R0) reports the total change in percentage points; arrows in R1–R3 indicate changes from the previous round.

Section 4.4 examines **feedback errors** by question type, difficulty, and ECG category/capability, with difficulty-level counts in Table 5.

4.3 Validation-Set Performance

Table 3 and Figure 4(c) summarize changes from R0 to R3 on MedX and LUDB. MedX accuracy improves from 51.57% to 57.24%, a **gain of 5.67 percentage points**, with a small R2 decline. LUDB accuracy increases across the recorded rounds from 53.45% through 61.28% and 63.28% to 67.24%, a **total gain of 13.79 percentage points**.

Figure 4 summarizes **round-wise benchmark and validation performance** together with historical strategy trajectories.

Table 3 Validation accuracy on MedX and LUDB across four rounds. The change columns report percentage-point differences between consecutive rounds and between R3 and R0. Both datasets improve overall; LUDB increases at every round, whereas MedX shows a small intermediate decline.

Dataset	Accuracy (%)				Change (pp)			
	R0	R1	R2	R3	R1–R0	R2–R1	R3–R2	R3–R0
MedX	51.57	55.94	55.86	57.24	+4.37	-0.08	+1.38	+5.67
LUDB	53.45	61.28	63.28	67.24	+7.83	+2.00	+3.96	+13.79

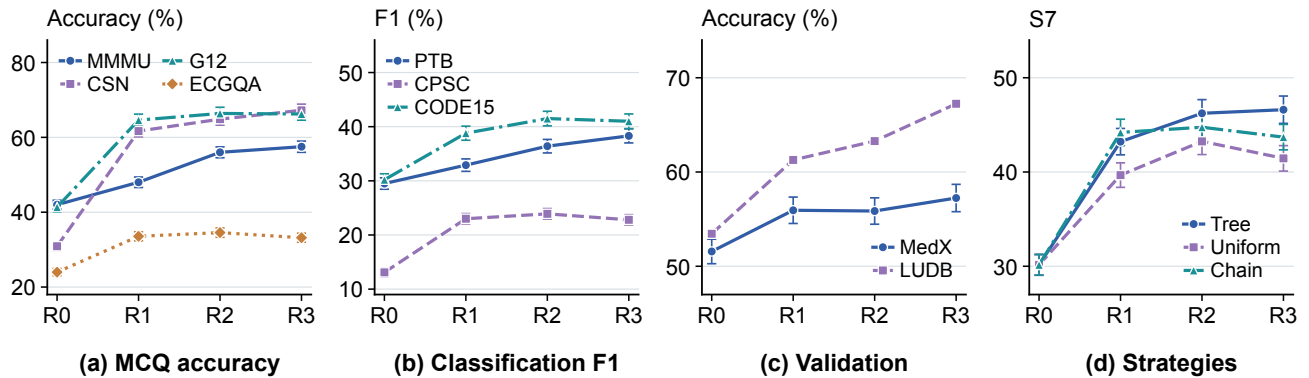


Figure 4 Round-wise benchmark, validation, and strategy results. (a) Four multiple-choice question (MCQ) accuracy metrics; (b) three classification F1 metrics; (c) MedX and LUDB validation accuracy; (d) seven-benchmark aggregate score (S7) for tree, uniform-allocation, and chain trajectories. Higher values indicate better performance. Error bars show ± 1 standard deviation (SD) where available; LUDB has no SD bars. The panels distinguish broad gains from task-specific fluctuations and the late-round declines of the two strategy controls.

4.4 Feedback Diagnostics across Rounds

The difficulty-stratified results distinguish aggregate progress from residual training needs. Error rates decrease from 26.11% to 12.97% for L1, from 38.52% to 24.43% for L2, and from 58.20% to 38.17% for L3. These correspond to reductions of 13.14, 14.09, and 20.03 percentage points. Despite the largest absolute reduction, **L3 retains the highest R3 error rate**. In particular, L3 error decreases by 6.39, 7.40, and 6.24 percentage points over successive updates, even as the corresponding **external S7 gains shrink** to 13.07, 3.01, and 0.38 points. Persistent improvement on difficult feedback questions therefore does not imply comparable gains on external tasks (Section 5.9).

Tables 4, 5, and 6 summarize the error rates by question type, difficulty, and ECG category, respectively. These are **development-feedback statistics** used to diagnose data needs, rather than independent benchmark scores. Overall feedback uses **3,000 questions**, with 1,140, 966, 811, and 688 errors at R0–R3. The difficulty groups reproduce these totals. The category breakdown remains unreconciled (Appendix B).

Table 4 Question-type feedback error rates (%). PM: perception/measurement; IA: interpretation/analysis; CD: clinical decision/treatment.

Round	PM	IA	CD
<i>N</i>	1,037	1,186	777
R0	32.98	38.87	43.37
R1	26.62↓	32.80↓	38.74↓
R2	<u>21.31</u> ↓	<u>27.57</u> ↓	<u>33.85</u> ↓
R3	17.65 ↓	23.10 ↓	29.73 ↓
$\Delta_{0 \rightarrow 3}$	15.33↓	15.77↓	13.64↓

Table 5 Difficulty-level feedback error rates (%). *N*: group size. $\Delta_{0 \rightarrow 3}$: R0–R3 reduction in percentage points. Full error counts appear in Table 19.

Round	L1	L2	L3
<i>N</i>	1,218	1,093	689
R0	26.11	38.52	58.20
R1	20.28↓	33.12↓	51.81↓
R2	<u>16.09</u> ↓	<u>28.27</u> ↓	<u>44.41</u> ↓
R3	12.97 ↓	24.43 ↓	38.17 ↓
$\Delta_{0 \rightarrow 3}$	13.14↓	14.09↓	20.03↓

Blue shading marks R3; bold/underlining mark the lowest/second-lowest rates. Blue arrows indicate decreases from the previous round; $\Delta_{0 \rightarrow 3}$ gives the total decrease in percentage points.

Table 6 Provisional ECG-category feedback error rates (%); category records await reconciliation with the overall reference. *N*: recorded sample count; $\Delta_{0 \rightarrow 3}$: R0–R3 reduction (pp, computed before rounding). Blue marks R3; bold/underline mark the best/second-best rates; ↓ marks reductions. Full names and counts: Table 20.

Round	<i>NSR</i>	<i>AF</i>	<i>AFL</i>	<i>ST</i>	<i>SB</i>	<i>PVC</i>	<i>PAC</i>	<i>AVB-I</i>	<i>Mobitz-I</i>	<i>Mobitz-II</i>	<i>AVB-III</i>	<i>LBBB</i>	<i>RBBB</i>	<i>Ant-STEMI</i>	<i>Inf-STEMI</i>	<i>Lat-STEMI</i>	<i>Post-STEMI</i>	<i>LVH</i>	<i>RVH</i>	<i>LQTS</i>	<i>HyperK</i>	<i>WPW</i>	<i>Brugada</i>
<i>N</i>	168	154	147	132	128	143	136	121	112	105	118	139	141	149	145	116	108	152	124	134	117	113	101
R0	25.0	39.6	44.9	28.0	26.6	40.6	39.0	38.0	43.8	48.6	44.1	36.7	35.5	44.3	43.4	46.6	49.1	37.5	41.1	41.0	42.7	41.6	45.5
R1	19.6↓	32.5↓	38.1↓	22.0↓	20.3↓	33.6↓	32.4↓	31.4↓	36.6↓	41.0↓	36.4↓	30.2↓	29.1↓	37.6↓	36.6↓	38.8↓	40.7↓	30.9↓	33.9↓	34.3↓	35.9↓	34.5↓	37.6↓
R2	<u>14.9</u> ↓	<u>26.6</u> ↓	<u>32.0</u> ↓	<u>16.7</u> ↓	<u>15.6</u> ↓	<u>27.3</u> ↓	<u>26.5</u> ↓	<u>25.6</u> ↓	<u>30.4</u> ↓	<u>34.3</u> ↓	<u>29.7</u> ↓	<u>24.5</u> ↓	<u>23.4</u> ↓	<u>31.5</u> ↓	<u>30.3</u> ↓	<u>31.9</u> ↓	<u>34.3</u> ↓	<u>25.0</u> ↓	<u>27.4</u> ↓	<u>28.4</u> ↓	<u>29.9</u> ↓	<u>28.3</u> ↓	<u>30.7</u> ↓
R3	11.3 ↓	21.4 ↓	26.5 ↓	12.9 ↓	11.7 ↓	21.7 ↓	21.3 ↓	19.8 ↓	24.1 ↓	27.6 ↓	23.7 ↓	19.4 ↓	18.4 ↓	26.2 ↓	24.8 ↓	25.9 ↓	27.8 ↓	19.7 ↓	21.8 ↓	22.4 ↓	23.9 ↓	22.1 ↓	23.8 ↓
$\Delta_{0 \rightarrow 3}$ ↓	13.7	18.2	18.4	15.2	14.8	18.9	17.6	18.2	19.6	21.0	20.3	17.3	17.0	18.1	18.6	20.7	21.3	17.8	19.4	18.7	18.8	19.5	21.8

5 Ablation Studies

We first assess **waveform validity** and the contribution of synthesis components. We then examine three linked questions: whether finer feedback improves **training utility**, whether the most prominent weakness determines the best training direction, and whether continued feedback improvement translates into external gains.

5.1 Effectiveness of the Synthesis Method

Quantitative waveform-feature validation. We report **waveform-feature error statistics**, including mean absolute error (MAE) and root mean squared error (RMSE), from a single synthesis run of 10,000 samples in Table 7.

Table 7 reports **errors separately** for HR, RR, PR, QRS, QT, and QTc.

All six features meet the adopted MAE criteria: HR, RR, PR, QRS, QT, and QTc. The reported MAEs support numerical fidelity.

RR also meets its criterion: its MAE is 15.28 ms, below the 20 ms limit. Average error alone cannot establish individual or clinical validity.

Table 7 Waveform-feature errors from a single synthesis run of 10,000 samples and comparison with the adopted MAE criteria. Blue marks MAE values meeting the adopted criteria.

Feature	Unit	Bias	MAE	RMSE	s_e^2	MAE limit	Status
HR	bpm	0.63	2.14	2.22	4.71	≤ 3	Met
RR	ms	3.11	15.28	15.36	238.62	< 20	Met
PR	ms	0.57	7.71	7.78	63.84	≤ 25	Met
QRS	ms	4.12	4.58	4.61	6.37	≤ 30	Met
QT	ms	0.94	9.71	9.85	101.43	≤ 30	Met
QTc	ms	0.86	10.22	10.34	114.08	≤ 40	Met

Error is remeasured minus target. Bias, MAE, RMSE, and limits use the listed unit; s_e^2 uses squared units and denominator $n - 1$. Limits are author-specified evaluation criteria, not confidence intervals or externally validated clinical standards.

For each feature, let $e_i = \hat{x}_i - x_i$ and compute

$$\bar{e} = \frac{1}{n} \sum_i e_i, \quad \text{MAE} = \frac{1}{n} \sum_i |e_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_i e_i^2}, \quad s_e^2 = \frac{1}{n-1} \sum_i (e_i - \bar{e})^2. \quad (4)$$

5.2 Blinded Review by Experienced Human Practitioners

We randomly sample 240 examples stratified by six edit types (HR, RR, PR, QRS, QT, and QTc), with 40 examples per type. **Two experienced human practitioners** score the examples **independently**. We report 95% confidence intervals (CIs) for the review metrics.

In stage 1, target labels, generation parameters, and teacher responses are hidden while waveform morphology and lead consistency are assessed. In stage 2, the practitioners inspect the before/after waveforms and editing targets to assess target attainment and non-target preservation. Stage 3 assesses answers, explanations, and expected-parameter references.

Answer correctness and evidence support.

Answer accuracy is 0.88, while explanation support is 0.76: a gap of 12 percentage points. Correct answers therefore merit a separate check of whether the stated reasoning is grounded in the waveform. Inter-rater agreement is $\kappa = 0.78$. The three review stages distinguish waveform morphology, editing fidelity, and the final answer with its explanation. Observed errors preclude treating automatic acceptance as a guarantee of correctness.

Table 8 Clinical review of 240 examples by two experienced human practitioners. Stages 1–3 assess morphology, editing fidelity, and answers/explanations, respectively. Values include reported 95% CIs.

Metric	Value	95% CI	n	Source
Target attainment	0.82	[0.76, 0.87]	240	Stage 2
Non-target preservation	0.79	[0.73, 0.84]	240	Stage 2
Morphology consistency	0.86	[0.81, 0.90]	240	Stage 1
Lead consistency	0.83	[0.78, 0.88]	240	Stage 1
Answer accuracy	0.88	[0.83, 0.92]	240	Stage 3
Explanation support	0.76	[0.70, 0.81]	240	Stage 3
Expected-parameter references	0.09	[0.06, 0.13]	240	Stage 3
Inter-rater κ	0.78	[0.71, 0.85]	240	Two raters

The reported CI methods are Wilson intervals or bootstrap; the metric-specific mapping and rater-aggregation rule are not specified in the available summary; κ denotes Cohen’s κ . Lower expected-parameter reference rates are better; higher values are better for the other proportion metrics.

Variation across edit types. Target attainment ranges from 0.78 for RR to 0.85 for HR. Explanation support varies from 0.72 to 0.78 and remains below answer accuracy for every edit type. The stratified results show that the overall gap is shared across editing targets rather than confined to one group.

The columns distinguish target attainment, preservation of non-target features, morphology consistency, answer accuracy, and explanation support. This stratification reveals variation across editing targets that is obscured by the overall averages.

Table 9 Clinical review by edit type: 40 examples per type, 240 overall. All values are proportions; higher is better.

Edit	<i>n</i>	Target	Preserve	Morph.	Answer	Support
HR	40	0.85	0.80	0.88	0.90	0.78
RR	40	0.78	0.75	0.82	0.85	0.72
PR	40	0.80	0.78	0.85	0.88	0.75
QRS	40	0.83	0.81	0.87	0.89	0.77
QT	40	0.82	0.79	0.86	0.87	0.76
QTc	40	0.84	0.80	0.88	0.89	0.78
Total	240	0.82	0.79	0.86	0.88	0.76

5.3 Multi-Model Question-Quality Scores

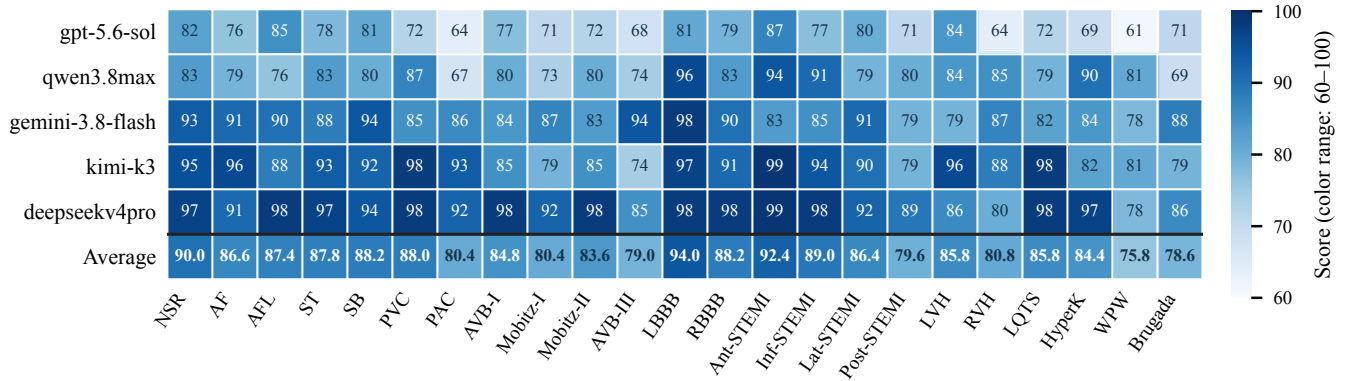


Figure 5 Recorded scores from five models for synthetic ECG questions across 23 categories (0–100 points). The bottom row is the per-category arithmetic mean across models; the shared color scale spans 60–100, from light blue (low) to dark blue (high). These scores are not diagnostic accuracy.

Figure 5 presents **recorded scores** from five models for synthetic ECG questions across 23 categories.

Question quality scoring and acceptance gates. Figure 5 summarizes quality scores from five models across 23 synthetic-question categories; the bottom row reports **per-category means**. R1 uses 10,000 questions from 16,193 candidates.

Interpreting model-based quality scores. These ratings measure the assessed **quality of constructed questions**, not the learner’s diagnostic performance. Agreement among model scores therefore complements, rather than replaces, **waveform remeasurement and review by experienced human practitioners**. The clinical review separately assesses target attainment, non-target preservation, answer correctness, and evidence support; downstream benchmarks assess training utility.

5.4 Data Construction and Feedback Selection

We compare four data configurations starting from the same checkpoint with $S7 = 43.22$. Each configuration adds 10,000 questions for continued SFT, using ordinary training-side questions, forward-only synthesis, forward and reverse synthesis with fixed selection proportions, or forward and reverse synthesis with **feedback-guided selection**. Each group adds 10,000 questions, and $\Delta S7$ is measured against that checkpoint. The comparisons separate ordinary questions, forward synthesis, added reverse synthesis, and feedback-guided selection. Bidirectional synthesis with feedback yields the **largest gain**.

Table 10 shows that ordinary questions and forward-only synthesis reach S7 scores of 44.15 and 44.27, respectively. Adding reverse synthesis with fixed selection proportions raises S7 to 45.56, **1.29 points** above forward-only synthesis; feedback-guided selection further raises it to 46.23, an additional **0.67 points**.

Table 10 Data-construction and selection ablation from a shared checkpoint with S7 = 43.22.

Group	Added data	S7	Δ S7
A	Ordinary training-side questions	44.15	+0.93
B	Forward-only synthesis	44.27	+1.05
C	Forward + reverse, fixed selection proportions	45.56	+2.34
D	Forward + reverse, feedback-guided selection	46.23	+3.01

5.5 Synthesis-Component Ablation

Table 11 compares synthesis updates at a fixed target of 10,000 accepted questions. The full method requires 16,193 candidates to reach this target, versus 36,723 with operator updates frozen and 22,398 with skill updates frozen (2.27 and 1.38 times as many, respectively). Freezing updates reduces question-quality scores and downstream R1 S7.

Table 11 R1 synthesis efficiency and training outcomes at a fixed target of 10,000 accepted questions. Candidate counts include rejected questions, and the pass rate measures the fraction accepted. Question scores assess generated-data quality, whereas S7 measures downstream model performance; Δ S7 is relative to the full method. Freezing operator or skill updates increases candidate requirements and reduces both quality scores and downstream performance.

Method	Candidates required	Pass (%)	Question score	S7	Δ S7
Full method	16,193	61.76	82.30	43.22	0.00
Freeze operator updates	36,723	27.23	71.90	39.29	-3.93
Freeze skill updates	22,398	44.65	79.20	40.15	-3.07

Each method trains on 10,000 accepted questions. Candidates include rejections; Δ S7 compares with the full method.

5.6 Feedback Granularity and Training Utility

To isolate the effect of **feedback resolution**, we compare four selection configurations from the **same parent checkpoint** (S7 = 43.22), each with 10,000 newly accepted questions. Table 12 varies the feedback dimensions used to select synthesized examples. Its fixed-mixture and four-dimensional endpoints correspond to the fixed-selection and feedback-guided settings in Table 10; these are shared endpoints, not independent experiments.

Category-only feedback improves S7 by just 0.04 points over fixed selection. Adding question type raises the score from 45.60 to 45.98 (+0.38 points), and incorporating difficulty and confusion pairs together raises it to 46.23 (+0.25 points). The finest configuration therefore gains **0.67 points** over fixed selection and **0.63 points** over category-only feedback at the same added-data volume. These comparisons support the value of **fine-grained feedback** for **training utility**.

Table 12 Feedback granularity under a common parent and a fixed budget of 10,000 added questions per configuration. Δ S7 is relative to the parent score of 43.22. The final column isolates the gain over fixed selection. These are downstream training outcomes, not measurements of waveform validity.

Feedback used for selection	S7	Δ S7	Gain over fixed (pp)
None: fixed mixture	45.56	+2.34	0.00
ECG category	45.60	+2.38	+0.04
Category + question type	45.98	+2.76	+0.42
Category + type + difficulty + confusion pairs	46.23	+3.01	+0.67

5.7 Round-Wise Branch Selection and Data Composition

The retained path is **A**→**B**→**B**: direct reinforcement wins the first round, whereas confusion contrast wins both later rounds. Table 13 reports all three same-parent candidates at each transition. Each candidate adds 10,000 accepted questions with matched historical replay and training settings. Selection uses S_{sel} on V_{sel} , distinct from external S7.

Table 13 Actual candidate compositions and branch choices across three rounds. A, B, and C denote direct reinforcement, confusion contrast, and difficulty/task curriculum. Exactly one branch is retained per round. Percentages describe overlapping attributes of each branch's 10,000 examples, not mutually exclusive slices. Pair percentages cover the listed pair families jointly.

Round	Branch	New-data composition	S_{sel}	Decision
R0→R1	A	AV block 45%; L3 50%; measurement questions 60%	42.66	✓ Retain
	B	Mobitz I/II and LBBB/RBBB contrasts 55%; L3 40%	40.51	Archive
	C	L2+L3 60%; clinical-reasoning questions 50%	38.27	Archive
R1→R2	A	Four regional STEMI categories 50%; L3 55%; image-localization questions 45%	44.12	Archive
	B	Anterior/lateral and inferior/posterior STEMI contrasts 60%; L3 45%	45.08	✓ Retain
	C	L3 70%; multistep-reasoning questions 55%	43.55	Archive
R2→R3	A	AF/AFL/PAC/PVC categories 45%; L3 50%; rhythm-reading questions 55%	45.90	Archive
	B	AF/AFL and PAC/PVC contrasts 55%; L3 45%	46.03	✓ Retain
	C	L3 65%; evidence-chain explanation questions 50%	44.71	Archive

AV: atrioventricular; LBBB/RBBB: left/right bundle branch block; STEMI: ST-segment elevation myocardial infarction; AF/AFL: atrial fibrillation/flutter; PAC/PVC: premature atrial/ventricular contraction.

From weak-category coverage to discriminative evidence. Category-specific error rates in the following analysis use the provisional breakdown and require item-level reconciliation; the difficulty-level rates agree with the overall reference. At R0, L3 error is 58.20%, and the recorded Mobitz II error is 48.57%. Branch A allocates 4,500 questions to the targeted atrioventricular-block categories and 5,000 to L3, with overlap permitted, and wins with $S_{\text{sel}} = 42.66$. After this update, L3 error falls to 51.81% and Mobitz II error to 40.95%, reductions of 6.39 and 7.62 percentage points, respectively. Although the parent also has a reported Mobitz I/II confusion error of 52%, the contrast-focused candidate scores 2.15 points below direct reinforcement. Early gains suggest improved **weak-category and difficult-question coverage**.

At R1, L3 error remains 51.81%, posterior STEMI error is 40.74%, and the reported anterior/lateral STEMI confusion error is 41%. The winning B recipe allocates **6,000** questions to regional STEMI contrasts and 4,500 to L3. Its selection score of 45.08 exceeds A by **0.96 points** and C by 1.53 points. At R2, L3 error remains 44.41%, with AF and AFL errors of 26.62% and 31.97%. B again wins, allocating **5,500** questions jointly to AF/AFL and PAC/PVC contrasts and 4,500 to L3. Its score of 46.03 exceeds A by 0.13 points and C by 1.32 points. The final B–A margin is descriptive. These outcomes distinguish **weakness identification** from **intervention selection**. A high category error rate indicates where performance is poor, but does not determine whether additional category coverage, contrastive evidence, or a curriculum change will produce the best update. In the two later rounds, the retained recipes emphasize evidence separating related diagnoses rather than simply adding more examples from a weak category.

Table 14 Data retained along the selected lineage and subsequent feedback outcomes. Counts are derived from the winning branch percentages and its 10,000 accepted questions. Category-specific error changes use the provisional category records; difficulty-level changes agree with the overall reference. Category/pair counts overlap with L3 counts. Error changes describe the full selected model update, including historical replay; they do not isolate the causal effect of an individual subset. Blue shading marks the final update; bold counts identify the targeted data, and blue values mark error reductions.

Update	Retained data	Feedback errors before → after
R0→R1 (A)	AV block: 4,500 (45%); L3: 5,000 (50%); measurement: 6,000 (60%)	L3: 58.20→51.81% (−6.39 pp); Mobitz II: 48.57→40.95% (−7.62 pp)
R1→R2 (B)	Regional STEMI contrasts: 6,000 (60%); L3: 4,500 (45%)	L3: 51.81→44.41% (−7.40 pp); posterior STEMI: 40.74→34.26% (−6.48 pp)
R2→R3 (B)	AF/AFL and PAC/PVC contrasts: 5,500 (55%); L3: 4,500 (45%)	L3: 44.41→38.17% (−6.24 pp); AF: 26.62→21.43% (−5.19 pp); AFL: 31.97→26.53% (−5.44 pp)

Table 14 relates each retained recipe to the subsequent feedback trajectory. In the two later rounds, 4,500 L3 questions accompany error reductions of 7.40 and 6.24 percentage points, respectively. The regional STEMI update also reduces posterior STEMI error by 6.48 points; the final update reduces AF and AFL category errors by 5.19 and 5.44 points. These are outcomes of the **full model updates**, including replay, rather than isolated effects of the listed subsets. The 5,500 final-round contrast questions jointly cover AF/AFL and PAC/PVC; category gains do

not establish **pair-specific confusion reductions**.

5.8 Comparison of RSI Evolution Strategies

Table 15 and Table 16 summarize the historical tree, uniform-allocation, and chain trajectories in R-A. The **tree trajectory continues improving**, finishing 5.16 and 2.91 points above the two controls and 1.86 above chain’s peak.

All three strategies improve over R0. Tree reaches $S7 = 46.61$ and is the only strategy to improve from R2 to R3 (+0.38 points); uniform and chain decline by 1.80 and 1.05 points. Chain leads R1; **tree finishes highest**.

Table 15 S7 trajectories in R-A. The last three columns report total gain, the final-round change, and the final gap to tree. Blue shading marks tree; changes are in percentage points.

Strategy	R0	R1	R2	R3	R3–R0	R3–R2	R3 vs. tree
Tree	30.15	43.22	46.23	46.61	+16.46	+0.38	0.00
Uniform	30.15	39.67	43.25	41.45	+11.30	-1.80	-5.16
Chain	30.15	44.20	44.75	43.70	+13.55	-1.05	-2.91

Table 16 Synthesis-update ablations and historical strategy trajectories (S7, higher is better). Left: R1 mean $\pm$ standard deviation; Δ is relative to the full method. Right: round-wise S7; $\Delta_{0 \rightarrow 3}$ is the total improvement. Blue/red arrows denote increases/decreases from the reference or previous round. Bold/underlining mark the best/second-best scores in each comparison. Blue shading highlights the full method and R3.

Synthesis updates (R1)			Historical strategies			
Method	S7 (mean $\pm$ SD)	Δ	Round	Tree	Uniform	Chain
Full method	43.22 $\pm$ 1.40	0.00	R0	30.15	30.15	30.15
Freeze operators	39.29 $\pm$ 1.30	-3.93↓	R1	<u>43.22↑</u>	39.67↑	44.20↑
Freeze skills	<u>40.15 $\pm$ 1.30</u>	-3.07↓	R2	46.23↑	43.25↑	<u>44.75↑</u>
Freezing operator or skill updates lowers R1 S7 by 3.93 or 3.07 points. See Table 11 for synthesis efficiency.			R3	46.61↑	41.45↓	<u>43.70↓</u>
			$\Delta_{0 \rightarrow 3}$	+16.46↑	+11.30↑	+13.55↑

5.9 Results Analysis

Fine-grained feedback improves training utility. With a common parent and 10,000 added questions, four-dimensional feedback reaches $S7 = 46.23$, versus 45.56 for fixed selection and 45.60 for category-only feedback (Table 12). These gains of **0.67** and **0.63 points** support resolving capability gaps beyond category frequency. They measure training utility; waveform remeasurement and clinical review separately assess supervision validity.

Error diagnosis does not determine the best intervention. The retained $A \rightarrow B \rightarrow B$ path separates identifying a weakness from choosing how to address it (Table 13). Direct reinforcement wins initially; confusion contrast wins later, scoring 45.08 versus 44.12 and 43.55 in $R1 \rightarrow R2$, and 46.03 versus 45.90 and 44.71 in $R2 \rightarrow R3$. The alternatives are direct reinforcement and curriculum adjustment, respectively. Thus, **feedback proposes intervention hypotheses; candidate comparison selects the update**. The final 0.13-point margin is descriptive, without establishing statistical significance.

Round-to-round evolution reselects how to address residual errors. Target content shifts from AV-block coverage and measurement to regional STEMI and then rhythm contrasts, while the intervention shifts from reinforcement to confusion-focused supervision. Each update inherits the selected checkpoint and replays historical data; renewed feedback redirects candidate construction. This trajectory is consistent with an update changing not only performance but also **which training evidence is most useful next**. However, learner state and recipe content change together, so the results do not isolate state-dependent utility or establish a universal coverage-to-discrimination progression. Comparing fixed-target interventions across $R0$ – $R2$ checkpoints would help distinguish these explanations.

Local correction has diminishing external returns. Feedback error rates fall by **5.80, 5.17, and 4.10 percentage points** over successive updates, while external S7 gains shrink to **13.07, 3.01, and 0.38 points** (Tables 20 and 2). ECGQA,

CPSC, and CODE15 regress in the final round despite fewer feedback errors. Later-round selection should therefore consider **transfer value and regressions**, not only residual-error frequency. The trend does not identify its cause or an optimal stopping point. Feedback guides proposals, V_{sel} selects candidates, and external benchmarks check transfer; local correction alone is insufficient evidence of broader progress.

6 Conclusion

ECG-RSI combines **verified bidirectional ECG synthesis** with feedback-driven model improvement. S7 increases from **30.15 to 46.61**, and MedXpert ECG validation accuracy rises from 51.57% to 57.24%. Ablations link operator and skill updates to efficiency and performance gains.

Beyond ECG synthesis, the retained $A \rightarrow B \rightarrow B$ trajectory suggests that RSI should **separate error diagnosis from intervention selection** and reconsider training directions after each update. Continued feedback-error reductions yield diminishing S7 gains, motivating external checks of **transfer and regression** rather than treating local correction as sufficient progress.

Declaration of AI Assistance. AI tools assisted with **language polishing**, formatting, and figure preparation.

Reproducibility Statement. The main text reports benchmark trajectories, feedback diagnostics, controlled ablations, blinded clinical review, and model-based question-quality scores. Supplementary materials provide the construction and training protocols, category-level records, prompts, and **complete case responses**.

References

- [1] Ruoyi Liu, Yuelin Bai, Xiang Yue, and Ping Zhang. Teach multimodal LLMs to comprehend electrocardiographic images. *arXiv preprint arXiv:2410.19008*, 2024.
- [2] Xiang Lan, Feng Wu, Kai He, Qinghao Zhao, Shenda Hong, and Mengling Feng. GEM: Empowering MLLM for grounded ECG understanding with time series and images. *arXiv preprint arXiv:2503.06073*, 2025.
- [3] Jiarui Jin, Haoyu Wang, Xingliang Wu, Xiaocheng Fang, Xiang Lan, Zihan Wang, Deyun Zhang, Bo Liu, Yingying Zhang, Xian Wu, Hongyan Li, and Shenda Hong. ECG-R1: protocol-guided and modality-agnostic MLLM for reliable ECG interpretation. *CoRR*, abs/2602.04279, 2026. doi: 10.48550/ARXIV.2602.04279. URL <https://doi.org/10.48550/arXiv.2602.04279>.
- [4] Jintai Chen, Kuanlun Liao, Kun Wei, Haochao Ying, Danny Z. Chen, and Jian Wu. ME-GAN: Learning panoptic electrocardio representations for multi-view ECG synthesis conditioned on heart diseases. *arXiv preprint arXiv:2207.10670*, 2022.
- [5] Yongfan Lai, Jiabo Chen, Deyun Zhang, Yue Wang, Shijia Geng, Hongyan Li, and Shenda Hong. DiffuSETS: 12-lead ECG generation conditioned on clinical text reports and patient-specific information. *arXiv preprint arXiv:2501.05932*, 2025.
- [6] Andrew Sellergren, Sahar Kazemzadeh, Tiam Jaroensri, et al. MedGemma Technical Report. *arXiv preprint arXiv:2507.05201*, 2025. doi: 10.48550/arXiv.2507.05201. URL <https://arxiv.org/abs/2507.05201>.
- [7] Haitao Li, Ziyu Li, Yiheng Mao, et al. anyECG-chat: A Generalist ECG-MLLM for Flexible ECG Input and Multi-Task Understanding. *CoRR*, abs/2506.00942, 2025. URL <https://arxiv.org/abs/2506.00942>.
- [8] Hung Manh Pham, Jialu Tang, Aaqib Saeed, and Dong Ma. Q-Heart: ECG Question Answering via Knowledge-Informed Multimodal LLMs. *CoRR*, abs/2505.06296, 2025. URL <https://arxiv.org/abs/2505.06296>.
- [9] Alexander Selivanov, Friederike Jungmann, Jan Kehrner, et al. ECG-LLM: Foundation Model for ECG-Based Cardiac Reasoning. *CoRR*, abs/2607.16323, 2026. URL <https://arxiv.org/abs/2607.16323>.
- [10] William Han, Tony Chen, Chaojing Duan, et al. ELF: A Family of Encoder-Free ECG-Language Models. *CoRR*, abs/2601.18798, 2026. URL <https://arxiv.org/abs/2601.18798>.
- [11] Deyun Zhang, Xiang Lan, Shijia Geng, et al. MEET: A Multimodal ECG Dataset from MIMIC-IV-ECG with Signals, Images, Features and Interpretations. *CoRR*, abs/2507.15255, 2025. URL <https://arxiv.org/abs/2507.15255>.
- [12] Xu Wang, Jiaju Kang, Puyu Han, et al. ECG-Expert-QA: A Benchmark for Evaluating Medical Large Language Models in Heart Disease Diagnosis. *CoRR*, abs/2502.17475, 2025. URL <https://arxiv.org/abs/2502.17475>.
- [13] William Han, Chaojing Duan, Zhepeng Cen, et al. Signal, Image, or Symbolic: Exploring the Best Input Representation for Electrocardiogram-Language Models Through a Unified Framework. *CoRR*, abs/2505.18847, 2025. URL <https://arxiv.org/abs/2505.18847>.
- [14] Jinning Liang, Mingcheng Zhu, and Tingting Zhu. ECG Mirage: Revealing and mitigating the underutilisation of ECGs in vision-language models for clinical prediction. *arXiv preprint arXiv:2609.21755*, 2026. URL <https://arxiv.org/abs/2609.21755>.
- [15] Jungwoo Oh, Hyunseung Chung, Junhee Lee, et al. ECG-Reasoning-Benchmark: A Benchmark for Evaluating Clinical Reasoning Capabilities in ECG Interpretation. *CoRR*, abs/2603.14326, 2026. URL <https://arxiv.org/abs/2603.14326>.

- [16] Maxwell A. Xu, Harish Haresamudram, Catherine W. Liu, et al. How Well Do Multimodal Models Reason on ECG Signals? *CoRR*, abs/2603.00312, 2026. URL <https://arxiv.org/abs/2603.00312>.
- [17] Nibras Soubh, Eva Rasenack, Helge Haarmann, et al. Performance of Vision-Enabled Large Language Models in Image-Based Electrocardiogram Interpretation: Exploratory Evaluation. *Journal of Medical Internet Research*, 28:e86692, 2026. doi: 10.2196/86692. URL <https://doi.org/10.2196/86692>.
- [18] Juan Miguel Lopez Alcaraz and Nils Strodthoff. Diffusion-based conditional ECG generation with structured state space models. *arXiv preprint arXiv:2301.08227*, 2023.
- [19] Sergey Skorik and Aram Avetisyan. SSSD-ECG-nle: New label embeddings with structured state-space models for ECG generation. *arXiv preprint arXiv:2407.11108*, 2024.
- [20] Hyunseung Chung, Jiho Kim, Joon-myung Kwon, Ki-Hyun Jeon, Min Sung Lee, and Edward Choi. Text-To-ECG: 12-lead electrocardiogram synthesis conditioned on clinical text reports. *arXiv preprint arXiv:2303.09395*, 2023.
- [21] Zhuoyi Huang, Nutan Sahoo, Anamika Kumari, Girish Kumar, Kexuan Cai, Shixing Cao, Yue Kang, Tian Xia, Somya Chatterjee, Nicholas Hausman, Aidan Jay, Eric S. Rosenthal, Soundar Srinivasan, Sadid Hasan, Alex Fedorov, and Sulaiman Vesal. High-fidelity synthetic ECG generation via mel-spectrogram informed diffusion training. *arXiv preprint arXiv:2510.05492*, 2025.
- [22] Vitalii Bondar, Serhii Semenov, Vira Babenko, and Dmytro Holovniak. FlowECG: Using flow matching to create a more efficient ECG signal generator. *arXiv preprint arXiv:2509.10491*, 2025.
- [23] Yongfan Lai, Bo Liu, Xinyan Guan, et al. ECGTwin: Personalized ECG Generation Using Controllable Diffusion Model. *CoRR*, abs/2508.02720, 2025. URL <https://arxiv.org/abs/2508.02720>.
- [24] Yuxin Lin, Jing Ma, Suyu Dong, et al. TransDiffECG: Semantically controllable ECG synthesis via transformer-based diffusion modeling. *Journal of Biomedical Informatics*, 172:104948, 2025. doi: 10.1016/j.jbi.2025.104948. URL <https://doi.org/10.1016/j.jbi.2025.104948>.
- [25] Jong-Hwan Jang, Yong-Yeon Jo, Sora Kang, et al. A novel XAI framework for explainable AI-ECG using generative counterfactual XAI (GCX). *Scientific Reports*, 15:23608, 2025. doi: 10.1038/s41598-025-08080-5. URL <https://doi.org/10.1038/s41598-025-08080-5>.
- [26] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL <https://arxiv.org/abs/2303.17651>.
- [27] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: language agents with verbal reinforcement learning. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023. URL http://papers.nips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html.
- [28] Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-Instruct: Aligning Language Models with Self-Generated Instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023. URL <https://arxiv.org/abs/2212.10560>.
- [29] Can Xu, Qingfeng Sun, Kai Zheng, Xiubo Geng, Pu Zhao, Jiazhan Feng, Chongyang Tao, Qingwei Lin, and Daxin Jiang. WizardLM: Empowering large pre-trained language models to follow complex instructions. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2304.12244>.
- [30] Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. STaR: Bootstrapping Reasoning With Reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL <https://arxiv.org/abs/2203.14465>.
- [31] Shidong Yang, Ziyu Ma, Tongwen Huang, Yiming Hu, Yong Wang, and Xiangxiang Chu. CoEvolve: Training LLM Agents via Agent-Data Mutual Evolution. *arXiv preprint arXiv:2604.15840*, 2026. URL <https://arxiv.org/abs/2604.15840>.
- [32] NeoHorse Team. NeoHorse-1: Towards Recursive Self-Improvement via Agentic Post-Training with Routing Harness. *arXiv preprint arXiv:2609.08183*, 2026. URL <https://arxiv.org/abs/2609.08183>.
- [33] Andrew Zhao, Yiran Wu, Yang Yue, et al. Absolute Zero: Reinforced Self-play Reasoning with Zero Data. *arXiv preprint arXiv:2505.03335*, 2025. URL <https://arxiv.org/abs/2505.03335>.
- [34] Yicheng He, Chengsong Huang, Zongxia Li, Jiaxin Huang, and Yonghui Yang. VisPlay: Self-Evolving Vision-Language Models from Images. *arXiv preprint arXiv:2511.15661*, 2025. URL <https://arxiv.org/abs/2511.15661>.
- [35] Fanqing Meng, Lingxiao Du, Qiguang Chen, Ziqi Zhao, Haocheng Lu, Mengkang Hu, and Michael Qizhe Shieh. Rsibench-data: Benchmarking data-centric research for recursive self-improvement. *CoRR*, abs/2607.25886, 2026. doi: 10.48550/ARXIV.2607.25886. URL <https://doi.org/10.48550/arXiv.2607.25886>.
- [36] Wei Liu, Siya Qi, Yali Du, and Yulan He. Self-Play Only Evolves When Self-Synthetic Pipeline Ensures Learnable Information Gain. *CoRR*, abs/2603.02218, 2026. URL <https://arxiv.org/abs/2603.02218>.
- [37] Min Zeng, Guanxin Tan, Libin Cen, et al. VISA: Agentic Self-Evolving Data Synthesis for Multimodal Instruction Following.

- CoRR, abs/2608.26013, 2026. URL <https://arxiv.org/abs/2608.26013>.
- [38] Shangjian Yin, Zehao Zhao, Kavosh Asadi, Rui Liu, Yuchen Lu, Shike Mei, Hang Cui, Luke Simon, Zhouxing Shi, and Hamed Firooz. Recursive Self-Improvement via On-Policy Distillation for Reasoning. *arXiv preprint arXiv:2609.30652*, 2026. URL <https://arxiv.org/abs/2609.30652>.
- [39] Michael Kirchhof, Eleonora Gualdoni, Andrew Szot, Khashayar Gatmiry, Aryo Lotfi, Abbas Kazerouni, Omar Attia, Sanjoy Chowdhury, and Alexander Toshev. RLTL;DR: Self-improvement by Internalizing Self-generated Feedback. *arXiv preprint arXiv:2609.37633*, 2026. URL <https://arxiv.org/abs/2609.37633>.
- [40] Eric Zelikman, Eliana Lorch, Lester Mackey, and Adam Tauman Kalai. Self-Taught Optimizer (STOP): Recursively Self-Improving Code Generation. In *Conference on Language Modeling (COLM)*, 2024. URL <https://arxiv.org/abs/2310.02304>.
- [41] Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents. *arXiv preprint arXiv:2505.22954*, 2025. URL <https://arxiv.org/abs/2505.22954>. Version 3, revised March 2026.
- [42] Zefeng Wang, Minxi Yan, Jinhe Bi, Sikuan Yan, Volker Tresp, and Yunpu Ma. MetaSkill-Evolve: Recursive Self-Improvement of LLM Agents via Two-Timescale Meta-Skill Evolution. *arXiv preprint arXiv:2607.05297*, 2026. URL <https://arxiv.org/abs/2607.05297>.
- [43] Tong Zheng, Xidong Wu, Zheng Zhang, Zhankui He, Chaoyi Zhang, Benjamin Coleman, Ruoqiao Wei, Di Bai, Haolin Liu, Rui Liu, Xue Wang, Yue Zhuan, Wang-Cheng Kang, Renkai Xiang, Heng Huang, Xinwu Cheng, and Yunsong Guo. Dream-RSI: Recursive Self-Improvement through Evolving Worlds. Technical report, Google and collaborating institutions, 2026. URL <https://dream-rsi.com/assets/dream-rsi.pdf>.
- [44] Shu Liu, Shubham Agarwal, Monishwaran Maheswaran, Mert Cemri, Zhifei Li, Qiuyang Mang, Ashwin Naren, Ethan Boneh, Audrey Cheng, Melissa Z. Pan, Alexander Du, Kurt Keutzer, Alvin Cheung, Alexandros G. Dimakis, Koushik Sen, Matei Zaharia, and Ion Stoica. EvoX: Meta-Evolution for Automated Discovery. *arXiv preprint arXiv:2602.23413*, 2026. URL <https://arxiv.org/abs/2602.23413>.
- [45] Cehao Yang, Xiaojun Wu, Xueyuan Lin, Chengjin Xu, Xuhui Jiang, Hui Xiong, and Jian Guo. Datafoundry: Evolving data preparators via recursive self-improvement. *CoRR*, abs/2608.29966, 2026. doi: 10.48550/ARXIV.2608.29966. URL <https://doi.org/10.48550/arXiv.2608.29966>.
- [46] Jinxin Hu, Hao Deng, Haibo Xing, Lingyu Mu, Muyu Zou, Weiqin Yang, Sirui Chen, Bohao Wang, Zhezheng Hao, Hao Zhang, Zulong Chen, Shizhun Wang, Yu Zhang, Xiaoyi Zeng, and Jiawei Chen. Astar: Learning to propose evolution directions for self-evolving industrial AI systems. *CoRR*, abs/2608.27287, 2026. doi: 10.48550/ARXIV.2608.27287. URL <https://doi.org/10.48550/arXiv.2608.27287>.
- [47] Siyu Yan, Yizhen Gao, Yilin Wang, et al. DataEvolver: Self-Evolving Multi-Agent Data Construction for Text-Rich Image Generation. *CoRR*, abs/2606.31537, 2026. URL <https://arxiv.org/abs/2606.31537>.
- [48] Jenny Zhang, Bingchen Zhao, Wannan Yang, et al. Hyperagents. *CoRR*, abs/2603.19461, 2026. URL <https://arxiv.org/abs/2603.19461>.
- [49] Peng Xia, Rujun Han, Zifeng Wang, et al. RRSI: Regularized Recursive Self-Improvement of Agent Harnesses. *CoRR*, abs/2609.24972, 2026. URL <https://arxiv.org/abs/2609.24972>.
- [50] Xiaoyin Chen, Canwen Xu, Yite Wang, et al. Learning to Self-Evolve. *CoRR*, abs/2603.18620, 2026. URL <https://arxiv.org/abs/2603.18620>.
- [51] Meiduo Chong, Shaolei Zhang, Ju Fan, and Xiaoyong Du. EvoOntology: A Self-Evolving Ontology Layer for Data Agents. *CoRR*, abs/2609.15779, 2026. URL <https://arxiv.org/abs/2609.15779>.
- [52] Ruoyu Wu, Shenfu Xie, Yinqian Sun, Haibo Tong, and Feifei Zhao. Mediskill-evo: Process-constrained self-evolution for evidence-grounded clinical interaction. *CoRR*, abs/2608.23397, 2026. doi: 10.48550/ARXIV.2608.23397. URL <https://doi.org/10.48550/arXiv.2608.23397>.
- [53] Kaiyuan Liu, Ziyuan Zhuang, Rongxiang Weng, and Jieping Ye. RecurSE: Bounded Recursive Self-Evaluation for LLM Rubric Judges. *arXiv preprint arXiv:2608.24231*, 2026. URL <https://arxiv.org/abs/2608.24231>.
- [54] Dhruv Srikanth, Bingchen Zhao, Dixing Xu, Yuxiang Wu, and Zhengyao Jiang. Recursive self-improvement of AI research agents. *arXiv preprint arXiv:2609.26457*, 2026. URL <https://arxiv.org/abs/2609.26457>.
- [55] Xinghong Fu, Aravinth Kulanthaivelu, and Yutaro Yamada. Self Improvement via Fast Tree-search. *arXiv preprint arXiv:2609.19526*, 2026. URL <https://arxiv.org/abs/2609.19526>.
- [56] Shuhan Xue, Jianyuan Zhong, Ziyuan Nan, Wenbin Li, Zhaochen Yu, Jinchao Ding, Qiang Gao, Pengyu Zhan, Yuntong Zhang, Tian Cheng, Zhenfei Yin, Yingcheng Wu, and Ling Yang. ScienceBuddy: Recursive-in-Recursive Self-Improvement for Interactive Scientific Agents. *arXiv preprint arXiv:2609.17523*, 2026. URL <https://arxiv.org/abs/2609.17523>.
- [57] Xilin Dang, Weilin Ruan, Xue Yang, Jinghao Wang, Xiaowei Hu, Jinpeng Li, and Pheng-Ann Heng. MedZERO: Self-Evolving Agents for Open-Ended Medical Reasoning Through Controlled Knowledge Accumulation. *arXiv preprint arXiv:2610.08327*, 2026. URL <https://arxiv.org/abs/2610.08327>.

- [58] Enoch Yin, Bin Liu, and Zhengling Qi. When Is Enough Enough in Self-Evolving LLM Systems? [arXiv preprint arXiv:2610.04756](https://arxiv.org/abs/2610.04756), 2026. URL <https://arxiv.org/abs/2610.04756>.
- [59] LASA Team, Weiwen Xu, Hou Pong Chan, Long Li, Mahani Aljunied, et al. Lingshu: A Generalist Foundation Model for Unified Multimodal Medical Understanding and Reasoning. [CoRR](https://dblp.org/rec/journals/corr/abs-2506-07044), abs/2506.07044, 2025. URL <https://dblp.org/rec/journals/corr/abs-2506-07044>.
- [60] Junying Chen, Ruyi Ouyang, Anningzhe Gao, Shunian Chen, Guiming Hardy Chen, Xidong Wang, Ruifei Zhang, Zhenyang Cai, Ke Ji, Guangjun Yu, Xiang Wan, and Benyou Wang. Huatuoogpt-vision, towards injecting medical visual knowledge into multimodal llms at scale. [CoRR](https://arxiv.org/abs/2406.19280), abs/2406.19280, 2024. doi: 10.48550/ARXIV.2406.19280. URL <https://doi.org/10.48550/ARXIV.2406.19280>.

Appendix Contents and Reading Guide

The entries below link to all supplementary sections and their starting pages.

Section	Contents	Page
A	Supplementary Method Details	19
	Data recipes, construction repair, recipe search, and implementation settings.	
A.1	Student-Conditioned Data Recipes	19
	Learner-conditioned recipe design and controlled difficulty probes.	
A.2	Construction Feedback and Versioned Repair	20
	Construction failure diagnosis and version-controlled generator updates.	
A.3	Waveform Construction and Validation Details	20
	Waveform editing operators and signal-level validation rules.	
A.4	Bounded Recipe Search within the Learning Loop	20
	Candidate recipe branching, evaluation, and model promotion.	
A.5	Training Configuration	21
	Optimizer, batching, distributed training, and frozen components.	
A.6	Parameter Extraction and Selection Details	21
	ECG measurement extraction, reliability, and parameter selection.	
B	Supplementary Experiments and Data	22
	Feedback counts, branch records, and waveform measurements.	
B.1	Difficulty-Level Feedback Counts	22
	Round-wise feedback counts for each difficulty level.	
B.2	Category-Level Feedback Counts	22
	Round-wise feedback counts across ECG task categories.	
B.3	Supplementary Branch-Evaluation Records	23
	Candidate branch evaluation and selection records.	
B.4	Supplementary Waveform Feature Data	23
	Source and edited waveform feature measurements.	
B.5	Multi-Seed Results on ECG Tasks	24
	Multi-seed evaluation results and variability across ECG tasks.	
C	ECG Task Taxonomy	24
	Medical grouping of the 23 ECG task categories.	
D	Stage-wise Prompt Templates	25
	Instructions for the construction, review, repair, and learning stages.	
D.1	Constrained waveform construction	25
	Waveform editing constraints and structured construction outputs.	
D.2	Evidence-grounded question and context generation	25
	Question and context generation grounded in waveform evidence.	
D.3	Teacher answer and evidence explanation	25
	Teacher answers and supporting evidence explanations.	
D.4	Independent quality and acceptance review	25
	Independent checks for question quality and acceptance.	

Section	Contents (continued)	Page
D.5	Construction failure and versioned repair Failure analysis and versioned generator repair.	26
D.6	Learner-feedback diagnosis and recipe proposal Learner error diagnosis and next-round recipe proposals.	26
D.7	Candidate comparison and model promotion Candidate comparison criteria and model promotion decisions.	26
E	Prompts and Case Studies Complete worked examples with prompts and model responses.	27
E.1	Case 1: PR prolongation PR-interval editing and the corresponding question-answer example.	27
E.2	Case 2: Heart-rate slowing Heart-rate editing and the corresponding question-answer example.	30

A Supplementary Method Details

A.1 Student-Conditioned Data Recipes

Equation (2) defines the joint recipe over category, difficulty, and supervision representation.

Here g is a task/capability category, d is learner-relative difficulty, and s is a supervision representation. These are task categories rather than additional input modalities. Each recipe also registers **retrieval/edit constraints, replay, and budgets**. The operator and prompt library ϕ determines which requests can be constructed. Selecting among fixed templates changes the recipe; editing a template or operator changes the library version. These interventions **must be evaluated separately**.

Difficulty estimation. With the current learner and decoding settings fixed, answer each training-side candidate independently K_i times and estimate difficulty from its fraction of correct responses:

$$\hat{p}_{it} = \frac{1}{K_i} \sum_{r=1}^{K_i} \mathbf{1}[\hat{y}_{ir} = y_i]. \quad (5)$$

Group candidates into **reliably solved, mixed-success, and unresolved** questions.

Supervision representations. A verified question can have (i) the **original teacher rationale**, (ii) an **evidence-expanded rationale** making required intermediate observations explicit, and (iii) an **error-targeted rationale** adding checks motivated by the student’s observed failure pattern. “Chain of thought (CoT)” denotes teacher-generated explanations used for supervision. Store token count under the student’s tokenizer, required evidence-step coverage, and the checks added; concise reasoning can be complete, and long reasoning can be redundant or unsupported. Evidence expansion must **preserve the image, question, options, answer, and clinical facts** in a paired comparison.

Difficulty belongs to the current learner. **Register the rollout budget**, concept-level scoring, position-counterbalancing policy, minimum valid-output rate, and binning thresholds before branch comparison. Approximate bins are reliably solved, mixed success, and unresolved; $0/K_i$ means unresolved under this probe, not intrinsically impossible. **Parsing/truncation remain separate flags**: report end-to-end correctness and correctness conditional on valid output.

ECG-specific supervision repair. For a **QRS-width error**, a candidate expanded target **identifies QRS onset/offset, converts grid distance to milliseconds, and checks the threshold**, using independently verified fiducials. If measurements are already correct but the label is wrong, add the relevant **cross-lead differential evidence** rather than a longer measurement explanation. These interventions are chosen from the observed error type and waveform evidence. Contrastive probes, scored outputs, and signal references supply observable support.

Figure 3 shows how aggregate feedback becomes candidate selection and generation requests for subsequent branch validation.

A.2 Construction Feedback and Versioned Repair

Acceptance also checks leakage and diversity.

Repair by failure type. Acceptance records cover target attainment, non-target drift, waveform continuity, lead consistency, morphology, context, answer uniqueness, and generation cost. **Target errors or boundary artifacts** trigger operator-parameter or implementation repairs; **missing morphology** triggers applicability or source-retrieval changes; **question ambiguity** triggers question and context template repairs; **unsupported explanations** trigger rationale-prompt revisions.

Reusable rules and version records. Each skill specifies applicability, permitted sources, operator order, parameter ranges, failure handling, and question rules. Updates record the triggering feedback, changes, proposer, old and new versions, validation results, and next-round scope. **Compare the original and revised versions** on the same source signals, requests, random seeds, and attempt budgets, and adopt revisions after **independent operator and question regression checks**.

A.3 Waveform Construction and Validation Details

Bidirectional synthesis and reverse perturbation. The **forward route** starts from a real training waveform, extracts reliable measurements and supported labels, renders the ECG, and constructs a question with compatible context and an evidence-linked answer. The **reverse route** starts from a target feature or contrast, selects an eligible training waveform, applies a constrained operator sequence, and re-extracts the resulting evidence before rendering and question construction.

For example, a **PR-interval contrast** can request a longer PR only when **P onset and QRS onset are measurable**. The edit must meet the requested interval range while **preserving protected heart rate and QRS morphology** within fixed tolerances. More complex disease patterns require their own verified morphology and clinical evidence. Both routes then use the same question, context, and explanation checks. Bidirectional synthesis determines how examples can be created; dual feedback determines how this construction is repaired and which examples to request next.

More data can come from (i) newly retrieved real waveforms in the permitted training pool, (ii) reuse or reweighting of accepted training questions, (iii) translating training-question distractors into proposed target-feature requests, (iv) controlled edits, and (v) audited training-side donors for complex morphology. **Feasibility, applicability, and the output evidence** must be checked before creating supervision.

For a source S and permitted donor set, construct $S' = T_\delta(S)$ and re-extract $u' = f(S')$. Store **target error** $||f_k(S') - f_k(S)|| - \Delta_k^{\text{request}}$ and **protected drift** $|f_j(S') - f_j(S)|$ separately. Check continuity at edit boundaries, behavior outside the edit mask, lead alignment, and class-specific morphology. A source and every donor must be **patient-disjoint** from feedback, selection, and independent evaluation data.

A.4 Bounded Recipe Search within the Learning Loop

The two-loop structure is retained. **Construction feedback** decides whether examples are valid and how to repair their generation. **Learner feedback** supplies both the demand profile and a delayed utility signal for choosing a recipe. Each round follows five registered steps:

1. **Score the accepted learner** on V_{fb} and probe eligible training candidates to update category, difficulty, and observable-error profiles.
2. **Propose a bounded set** $\mathcal{A}_t$ of recipes using those profiles and prior branch outcomes; save the proposer identity and exact recipe.
3. **Construct and validate** each recipe’s training data under fixed gates. From the same parent checkpoint, run matched small-budget SFT branches with fixed replay and optimizer settings.
4. **Compare candidate learners** on the predesignated selection-development protocol, including forgetting and output validity.
5. **Promote the accepted candidate** and its associated recipe/policy versions, then remeasure the learner for the next round. If a pilot winner is retrained at full budget, restart from the registered parent and validate that full-budget checkpoint before promotion.

For branch a , the simplest candidate utility is

$$\Delta U_t(a) = U_{\text{sel}}(\text{SFT}(\theta_t, D_t^a \cup R_t; b)) - U_{\text{sel}}(\theta_t), \quad (6)$$

where b is the common pilot training budget and U_{sel} is a preregistered capability-macro score on a fixed eligible task set.

Each round trains candidate branches from the same parent, compares them on a fixed development set, and retains the **best qualifying branch**. The next round starts from that model, while unselected branches are archived. The search tree records candidates; the retained path records the selected model updates.

Figure 3 is a schematic of candidate exploration and checkpoint inheritance. Its colored path illustrates a possible lineage. Section 3.3.2 specifies how the three recipe families change the data; Table 13 gives the measured compositions, scores, and selections.

A.5 Training Configuration

Each round uses **three training epochs** and a maximum sequence length of 28,672 tokens. Packing, padding-free batching, and sequence parallelism are enabled. The micro-batch size is per GPU.

The **parallelism factors** use DP (data), TP (tensor), PP (pipeline), CP (context), EP (expert), and ETP (expert tensor). Training runs on 16 nodes with eight GPUs each.

Distributed Adam uses cosine learning-rate (LR) decay with 5% warmup. Computation uses bfloat16; master parameters, gradients, and Adam states use float32. Optimizer CPU offload is enabled with fraction 1.0.

The language model is trainable; the vision encoder and multimodal aligner are frozen. Selected modules are recomputed, with FlashAttention as the attention backend. GEMM denotes general matrix multiplication.

Table 18 Training hyperparameters and distributed settings.

Parameter	Setting
<i>Sequence and topology</i>	
Epochs / maximum tokens	3 / 28,672
Micro-batch / global batch	1 / 32
Gradient accumulation	1
GPUs (nodes $\times$ GPUs)	128 (16 $\times$ 8)
DP / TP / PP / CP	32 / 4 / 1 / 1
EP / ETP	8 / 1
<i>Optimization</i>	
Optimizer	Distributed Adam
Peak / minimum LR	10^{-5} / 10^{-7}
Schedule / warmup	Cosine / 5%
Weight decay / grad. clip	0.1 (constant) / 1.0
Adam $\beta_1/\beta_2/\epsilon$	0.9 / 0.95 / 10^{-8}
Compute / master precision	bfloat16 / float32
CPU offload fraction	1.0
<i>Modules and experts</i>	
Attention backend	FlashAttention
Recompute modules	core_attn, moe, mlp
Grouped GEMM	Enabled
Permutation fusion	Disabled
Shared-expert overlap	Enabled
Auxiliary loss / router dtype	10^{-6} / float32

A.6 Parameter Extraction and Selection Details

A delineator supplies candidate measurements

$$u = f(S), \quad u \in \mathbb{R}^M. \quad (7)$$

At 25 mm/s, a 1 mm horizontal square is 40 ms and a 5 mm square is 200 ms; at 10 mm/mV, 1 mm vertically is 0.1 mV (a 5 mm square is 0.5 mV). PR is measured from **P onset to QRS onset**, and QT from **QRS onset to T offset**.

The parameter selector uses **reliability and diagnostic utility**,

$$r_k = \frac{1/\hat{\sigma}_k}{\sum_j 1/\hat{\sigma}_j}, \quad (8)$$

$$s_{ik} = a_{ik} r_k d_k, \quad (9)$$

$$C_i = \{k : s_{ik} > \tau_{\text{param}}\}. \quad (10)$$

Here a_{ik} is definability, $\hat{\sigma}_k$ an extraction-error scale, and d_k a probe-based utility.

Scale specification. Comparing inverse error scales across features requires a **common dimensionless normalization**: errors measured in bpm, ms, and mV **cannot be combined directly without reference scales**. The normalization scales used in the implementation are not specified here and must be documented before these weights can be reproduced.

B Supplementary Experiments and Data

This appendix summarizes round-wise feedback statistics, branch-evaluation results, and waveform data. Overall feedback statistics use a **fixed set of 3,000 questions**, with 1,140, 966, 811, and 688 errors at R0–R3, respectively.

B.1 Difficulty-Level Feedback Counts

Residual difficulty. L3 records the largest absolute reduction, 20.03 percentage points, yet its final error rate remains 38.17%. L1 falls to 12.97%. Counts and group sizes distinguish changes in error rate from differences in group size.

For this table and Table 20, blue marks R3. Bold and underlined values identify the best and second-best rates; ↓ marks a decrease from the previous round.

Table 19 Difficulty-level errors: count (%). Group sizes appear in the header.

Round	L1 N = 1,218	L2 N = 1,093	L3 N = 689
R0	318 (26.11)	421 (38.52)	401 (58.20)
R1	247 (20.28↓)	362 (33.12↓)	357 (51.81↓)
R2	196 (16.09↓)	309 (28.27↓)	306 (44.41↓)
R3	158 (12.97↓)	267 (24.43↓)	263 (38.17↓)

B.2 Category-Level Feedback Counts

Tables 4, 5, and 6 in Section 4.4 summarize the feedback trajectories. The recorded category-level breakdown follows. These rows have not yet been reconciled with the overall reference counts and **must not be treated as an additive decomposition** of the 3,000-question set. **Category-specific comparisons** remain provisional pending item-level reconciliation.

Table 20 Recorded error counts and rates (%) for 23 ECG categories across R0–R3. Category rows are provisional; the separate overall reference row governs aggregate results. N: recorded group size.

Group	N	Error count				Error rate (%)			
		R0	R1	R2	R3	R0	R1	R2	R3
Normal Sinus Rhythm	168	42	33	25	19	25.00	19.64↓	14.88↓	11.31↓
Atrial Fibrillation	154	61	50	41	33	39.61	32.47↓	26.62↓	21.43↓
Atrial Flutter	147	66	56	47	39	44.90	38.10↓	31.97↓	26.53↓
Sinus Tachycardia	132	37	29	22	17	28.03	21.97↓	16.67↓	12.88↓
Sinus Bradycardia	128	34	26	20	15	26.56	20.31↓	15.63↓	11.72↓
Premature Ventricular Contraction	143	58	48	39	31	40.56	33.57↓	27.27↓	21.68↓
Premature Atrial Contraction	136	53	44	36	29	38.97	32.35↓	26.47↓	21.32↓
First-Degree AV Block	121	46	38	31	24	38.02	31.40↓	25.62↓	19.83↓
Mobitz I AV Block	112	49	41	34	27	43.75	36.61↓	30.36↓	24.11↓

Group	N	Error count				Error rate (%)			
		R0	R1	R2	R3	R0	R1	R2	R3
Mobitz II AV Block	105	51	43	36	29	48.57	40.95↓	34.29↓	27.62↓
Complete AV Block	118	52	43	35	28	44.07	36.44↓	29.66↓	23.73↓
Left Bundle Branch Block	139	51	42	34	27	36.69	30.22↓	24.46↓	19.42↓
Right Bundle Branch Block	141	50	41	33	26	35.46	29.08↓	23.40↓	18.44↓
Anterior STEMI	149	66	56	47	39	44.30	37.58↓	31.54↓	26.17↓
Inferior STEMI	145	63	53	44	36	43.45	36.55↓	30.34↓	24.83↓
Lateral STEMI	116	54	45	37	30	46.55	38.79↓	31.90↓	25.86↓
Posterior STEMI	108	53	44	37	30	49.07	40.74↓	34.26↓	27.78↓
Left Ventricular Hypertrophy	152	57	47	38	30	37.50	30.92↓	25.00↓	19.74↓
Right Ventricular Hypertrophy	124	51	42	34	27	41.13	33.87↓	27.42↓	21.77↓
Long QT Syndrome	134	55	46	38	30	41.04	34.33↓	28.36↓	22.39↓
Hyperkalemia	117	50	42	35	28	42.74	35.90↓	29.91↓	23.93↓
Wolff-Parkinson-White Syndrome	113	47	39	32	25	41.59	34.51↓	28.32↓	22.12↓
Brugada Syndrome	101	46	38	31	24	45.54	37.62↓	30.69↓	23.76↓
Overall reference	3,000	1,140	966	811	688	38.00	32.20↓	27.03↓	22.93↓

B.3 Supplementary Branch-Evaluation Records

The parent and three candidates are compared on the same R0→R1 development protocol. The table reports overall accuracy, three macro scores, and selection score S ; ΔS is relative to the parent.

All candidates pass the score and decline gates. Branch A is promoted with the **highest selection score, 42.66**; B and C remain eligible but rank below A. Worst-group changes are reported in percentage points.

Table 13 gives all three rounds of candidate selection and the full A→B→B retained path.

Table 21 R0→R1 development scores and branch selection. Blue marks the promoted branch.

Metric	Parent	A	B	C
Overall (%)	30.15	43.22	41.09	38.78
Five-group macro (%)	29.80	42.10	40.25	38.10
Task-type macro (%)	30.05	42.85	40.80	38.45
Difficulty macro (%)	29.60	41.90	39.95	37.70
S	29.93	42.66	40.51	38.27
ΔS	–	+12.73	+10.58	+8.34
Worst-group change (pp)	–	-1.20	-1.80	-2.10
Score / decline gate	–	✓ Pass	✓ Pass	✓ Pass

B.4 Supplementary Waveform Feature Data

Table 22 Supplementary waveform-feature statistics.

Feature	Unit	Sample count	MAE	Mean bias	Target attainment (%)	Non-target drift
HR	bpm	HR	bpm	2,486	5.83	-0.47
RR	ms	RR	ms	2,147	12.46	1.38
PR	ms	PR	ms	1,324	14.72	-2.19
QRS	ms	QRS	ms	1,289	9.35	0.86
QT	ms	QT	ms	1,568	17.24	-1.73
QTc	ms	QTc	ms	1,186	19.48	2.05

B.5 Multi-Seed Results on ECG Tasks

Scores are reported for **three independent training runs**; SD is the sample standard deviation. The base row uses the same fixed initialization.

These scores **must not be interpreted as uncertainty estimates for the main S7 trajectory** until their evaluation protocol is identified. Mean ECG-task scores rise from 33.40 at R1 to 37.82 at R3, with SD between 0.22 and 0.29 across trained rounds.

Table 23 Reported three-seed ECG-task scores; their metric and evaluation-set correspondence to the main S7 results remain unspecified.

Model	Seed 1	Seed 2	Seed 3	Mean $\pm$ SD
Base	30.15	30.15	30.15	30.15 ± 0.00
ECG-RSI R1	33.42	33.18	33.61	33.40 ± 0.22
ECG-RSI R2	35.27	35.04	35.48	35.26 ± 0.22
ECG-RSI R3	37.86	37.52	38.09	37.82 ± 0.29

C ECG Task Taxonomy

Figure 6 shows the 23 ECG tasks currently used in this paper. Tasks are organized by medical category. Sector area represents the number of categories, not training-sample counts or model scores.

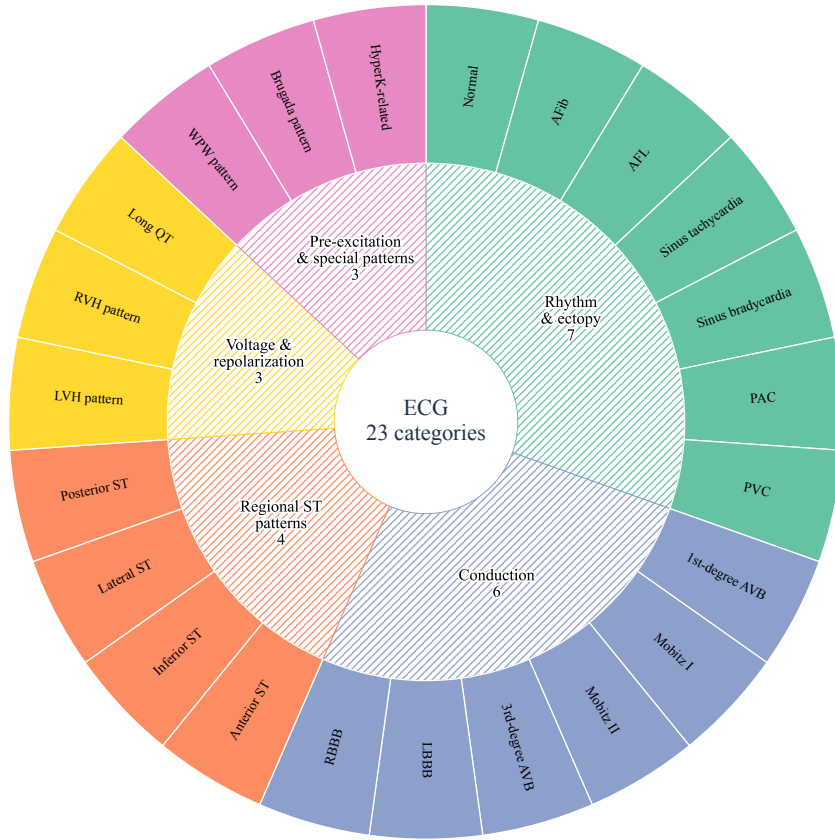


Figure 6 Medical taxonomy organizing the 23 ECG task categories. Each outer sector denotes one category; inner sectors group related categories, with areas and counts indicating the number of categories in each group. The diagram describes task coverage and the grouping used to interpret category-level feedback, rather than the frequency of examples in the training data.

D Stage-wise Prompt Templates

This appendix documents reusable prompt templates for the method interfaces. Bracketed fields are filled with request-specific inputs and tool results. Section D.3 reproduces the teacher-answer instruction; the other templates are organized by function. The following cases retain their original questions and saved responses.

D.1 Constrained waveform construction

Prompt P1

System role. You are an ECG waveform-construction agent. Convert a target feature request into an auditable edit plan.

User inputs. [request ID], [training-source waveform and sampling rate], [lead names and calibration], [measured landmarks and uncertainty], [target ranges], [protected features and tolerances], [allowed operators and donors], [retry budget].

Instructions. **Check source eligibility and landmark reliability.** For the **forward route**, extract supported features without changing the signal. For the **reverse route**, specify operator order, parameters, and edit mask. Invoke the supplied editing and measurement tools, then **compare observed targets and protected drift** with the fixed tolerances. Check boundary continuity, lead alignment, and required morphology. If tools or required landmarks are unavailable, return blocked; if verification fails, repair within budget or reject.

Output fields. Status; source identities; operator plan; requested and observed measurements with units; protected drift; validity checks; failure codes; edited-signal reference. **Report executed results separately from proposed operations.**

D.2 Evidence-grounded question and context generation

Prompt P2

System role. You construct ECG multiple-choice questions from verified signal evidence. Keep internal labels and generator intent out of the student-visible question.

User inputs. [verified ECG image], [measurement evidence], [supported label level], [target category], [question type and difficulty], [option count], [context policy], [permitted source context].

Instructions. Write one question with exactly the requested number of options and **one evidence-supported answer**. Use plausible distractors that remain incorrect for this signal. Ask only at the supported level: measurement, ECG pattern, or clinically supported diagnosis. Default to image-only or traceable demographics. **Mark any synthetic teaching context explicitly** and check it against the achieved waveform.

Output fields. Student-visible question, options, and allowed context; evaluator-only answer, evidence mapping, context provenance, category, and difficulty; ambiguity or insufficient-evidence flag.

D.3 Teacher answer and evidence explanation

Prompt P3

Please reason step by step, and put your final answer within `\boxed{}`.

D.4 Independent quality and acceptance review

Prompt P4

System role. You audit a candidate ECG training example.

User inputs. [source and edited evidence], [target/protected-feature checks], [question and options], [context provenance], [proposed answer and teacher response], [fixed rubric and aggregation rule].

Instructions. **Independently solve the question**, then audit target attainment, protected drift, morphology, context consistency, answer uniqueness, leakage, and explanation support. Return separate decisions for each gate and cite the relevant observation. Score only with the supplied rubric. The aggregate-score threshold is **80/100**, but **every mandatory**

gate must also pass.

Output fields. Independent answer; gate-wise pass/fail decisions and evidence; rubric scores; aggregate if defined; accept/review/reject status; failure codes and repair target.

D.5 Construction failure and versioned repair

Prompt P5

System role. You propose a minimal repair to a failed data-construction component while keeping the acceptance criteria fixed.

User inputs. [failed attempt and evidence], [failure codes], [current operator/prompt/skill version], [allowed repair scope], [fixed regression suite], [attempt budget].

Instructions. Map numerical or boundary failures to operator repair; missing morphology to applicability or donor retrieval; ambiguity/leakage to question or context repair; unsupported explanation to response-prompt repair. **Specify the exact change** and why it addresses the observed failure. **Compare old and new versions** on the same sources, requests, seeds, and budget. Quarantine or reject when the budget is exhausted.

Output fields. Triggering failure; component; exact revision; old/new version; proposer; validation results; acceptance rationale; next-round scope.

D.6 Learner-feedback diagnosis and recipe proposal

Prompt P6

System role. You propose training-data interventions from development feedback.

User inputs. [accepted parent and feedback version], [category/type/difficulty counts and errors], [confusions and regressions], [training-pool coverage], [construction yield/cost], [history], [token and attempt budgets].

Instructions. **Prioritize deficits with adequate support**; report sparse or uncertain groups explicitly. Propose distinct candidates for **direct weakness coverage**, **confusion contrasts**, and **difficulty or evidence-granularity progression**. For each, give normalized category, difficulty, and supervision-style allocations; requested feature ranges; protected attributes; allowed routes/operators; source filters; and cost limits. A/B/C retain the same previous cumulative training set unchanged and each add 10,000 first-use accepted questions, without an extra shared baseline set; **only the winning additions enter the next cumulative set**.

Output fields. candidate recipes and hypotheses; normalized allocations and budgets; construction requests; provenance restrictions.

D.7 Candidate comparison and model promotion

Prompt P7

System role. You apply a preregistered development-only promotion rule to comparable candidate learners.

User inputs. [parent and candidate checkpoint IDs], [new/replay manifests], [supervised-token budgets], [fixed selection-set results], [primary score], [regression tolerances], [tie/stopping rules].

Instructions. **Check that candidates share the required parent**, training budget, and evaluation protocol. Apply the supplied score, regression, and valid-output gates exactly; do not invent thresholds. **Promote only an eligible candidate**, otherwise retain the parent. Count only previously unused accepted items from the promoted branch toward cumulative additions, preserving replay provenance. **Archive other branches and log the decision**.

Output fields. Comparability checks; gate results; chosen checkpoint or retain-parent decision; supporting scores; admitted item manifest; replay manifest; stopping status.

Case 1 — PR-interval prolongation

Waveform source: PTB-XL - train split

(a) Original waveform

(b) Perturbed waveform

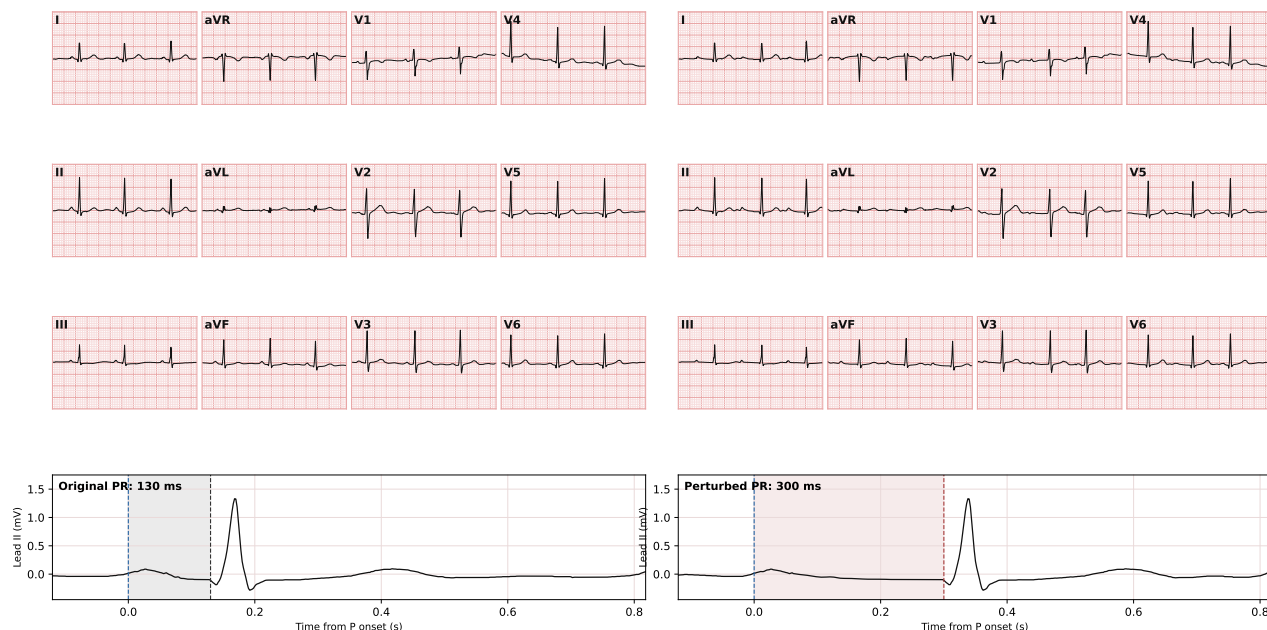


Figure 7 Targeted PR prolongation, shown through original and edited ECGs with lead II details. The edit lengthens the PQ segment and compensates in the TP segment to preserve the RR interval. **The target PR is 300 ms** and the remeasured PR is 298 ms; these automatic measurements support the subsequent parameter-comparison question.

E Prompts and Case Studies

Reading the saved responses. The following transcripts retain their recorded model label, Qwen3.8; they should not be attributed to the Qwen3.5-35B-A3B learner without model-provenance records. A correct final option does not establish that every intermediate observation is supported. These are construction examples with unedited responses, not demonstrations that every explanation passes the evidence checks. **Editorial emphasis only:** **blue bold text** marks reasoning cues and reported values; **amber text** marks assumptions or grounding concerns. The transcript wording is unchanged.

E.1 Case 1: PR prolongation

Case 1: PR prolongation

Waveform construction

Copy the P wave and QRS–T segment, lengthen the PQ segment, and shorten the TP segment to preserve the RR interval. **The target PR is 300 ms**; remeasurement yields **PR 298 ms** and **HR 74.6 bpm**.

Question

Question: Under the automatic closed-loop measurement convention of this data-construction pipeline, which parameter analysis and conclusion are correct for this 12-lead ECG?

Options:

A: **HR 74.6 bpm**, PR 130.0 ms, QRS 96.0 ms, QTc 393.7 ms -> Conclusion: no first-degree atrioventricular block

B: HR 48.0 bpm, **PR 298.0 ms**, QRS 96.0 ms, QTc 393.7 ms -> Conclusion: sinus bradycardia with first-degree atrioventricular block

C: **HR 74.6 bpm**, **PR 298.0 ms**, QRS 96.0 ms, QTc 393.7 ms -> Conclusion: first-degree atrioventricular block

D: **HR 74.6 bpm**, **PR 298.0 ms**, QRS 130.0 ms, QTc 393.7 ms -> Conclusion: first-degree atrioventricular block with intraventricular conduction delay

Please reason step by step, and put your final answer within `\boxed{}`.

Qwen3.8 response

Let me analyze this ECG question.

The question is asking (in Chinese): "According to the automatic closed-loop measurement criteria of this data construction pipeline, looking at this 12-lead ECG, which of the following parameter analyses and conclusions is correct?"

Options:

A: **HR 74.6 bpm**, PR 130.0 ms, QRS 96.0 ms, QTc 393.7 ms → Conclusion: No first-degree AV block observed

B: HR 48.0 bpm, **PR 298.0 ms**, QRS 96.0 ms, QTc 393.7 ms → Conclusion: Sinus bradycardia with first-degree AV block

C: **HR 74.6 bpm**, **PR 298.0 ms**, QRS 96.0 ms, QTc 393.7 ms → Conclusion: First-degree AV block

D: **HR 74.6 bpm**, **PR 298.0 ms**, QRS 130.0 ms, QTc 393.7 ms → Conclusion: First-degree AV block with intraventricular conduction delay

I need to look at the ECG and determine the parameters.

Looking at the ECG: the rhythm strip (II rhythm) shows about 12 beats over some length. Let me estimate the heart rate. The rhythm strip is probably 10 seconds? Let me count: I see 12 QRS complexes in the II rhythm strip. If the strip is 10 seconds, $HR \approx 12 \times 6 = 72$ bpm... actually 12 beats in 10 seconds = 72 bpm. Hmm, but the option says 74.6 bpm. That's plausible.

Looking at the PR interval: in lead II, there's a P wave before each QRS. The PR interval looks... let me see. In the rhythm strip, the P wave appears just before the QRS? Actually, there's a small bump (P wave) before the QRS, and a T wave after the QRS. Let me look more carefully.

In lead II (second row first column), there's a small P wave, then QRS, then T wave. The distance from P to QRS looks short? Hmm.

But the question says "perturbed waveform"—this is a synthetic case with perturbations added. The question is about the "automatic closed-loop measurement criteria" of the data construction pipeline. **Probably the ground truth parameters are:** HR 74.6, **PR 298 ms** (prolonged → first-degree AV block), QRS 96 ms (normal), QTc 393.7 ms.

Looking at the waveform: in lead II, before each QRS there's a P wave... hmm, actually looking at the rhythm strip: after each QRS there's a T wave, and before the next QRS there's a P wave? Let me look: pattern per beat: QRS spike, then T wave bump, then another bump (P wave?), then QRS. Actually, in the rhythm strip I see: spike, then bump, then bump, then spike... spacing: after the spike at $x=290$, bump at ~ 340 , bump at ~ 390 , spike at ~ 455 . Hmm, so two bumps between spikes: T wave and P wave. If the P wave is at ~ 390 and QRS at ~ 455 , then PR interval is ~ 65 px? **Need scale.**

Scale: 25 mm/s. Grid: **large box = 5 mm = 0.2 s = 200 ms.** **Small box = 1 mm = 40 ms.**

Let me estimate pixel scale. Each lead panel width is ~ 500 px? Hmm, need to guess from image coordinates. The image is 2338 px wide? Actually the given image dimensions: width ~ 2338 ? Let me think: the figure shows 4 columns of panels. Each panel ~ 500 px wide representing 10 seconds? No, each panel usually shows ~ 2.5 seconds? Hmm, but the rhythm strip at the bottom shows 12 beats over full width.

Better approach: use RR interval to determine HR. If HR is 74.6 bpm, $RR = 804$ ms. If HR is 48, $RR = 1250$ ms.

Looking at the rhythm strip: 12 beats over full width. If full width is 10 seconds (standard), then average $RR = 10/11$? Actually 12 beats → 11 intervals over ~ 10 s → $RR \approx 0.909$ s → $HR \approx 66$. Hmm.

But maybe the strip is longer? Let me count beats in rhythm strip: spikes at approx x positions: 290, 455, 610, 765, 920, 1075, 1225, 1380, 1530, 1680, 1835, 1990? That's 12 spikes. Spacing ~ 155 px. If spacing = 804 ms (HR 74.6), then px per ms = $155/804 \approx 0.1928$ px/ms → 25 mm/s = 0.025 mm/ms... hmm, let me compute mm: at 25 mm/s, 804 ms = 20.1 mm. So 155 px = 20.1 mm → 7.7 px/mm. Large box (5 mm) = 38.5 px. Small box = 7.7 px.

Now PR interval: P wave bump before QRS: in rhythm strip, bump at ~ 390 and spike at ~ 455 : difference 65 px = $65/7.7 \approx 8.4$ mm = 337 ms? Hmm, that's too long. But maybe the bump at 390 is the P wave and the bump at 340 is the T wave. $PR = 455 - 390 = 65$ px → at 7.7 px/mm → 8.44 mm → at 25 mm/s → 337 ms. Hmm, option says **PR 298 ms**. 298 ms = 7.45 mm = 57 px. Close to 65 px? **My pixel estimates** are rough.

Alternatively, if HR were 48 ($RR = 1250$ ms), spacing 155 px → $155/1250 = 0.124$ px/ms → px per mm = $0.124 \times 40 = 4.96$ px/mm. Then $PR = 65$ px = 13.1 mm = 524 ms, unlikely.

So HR ~ 74.6 is consistent with 12 beats over ~ 9.6 seconds? 11 intervals $\times 0.804$ s = 8.84 s. Rhythm strip width ~ 1980 px? Hmm, from $x=170$ to $x=2150 \approx 1980$ px. At 7.7 px/mm → 257 mm → at 25 mm/s → 10.3 s. OK, plausible: 10-second rhythm strip with 12 beats → HR ~ 72 -75. Good.

Case 1: PR prolongation (continued)

Now PR: is it prolonged (298 ms)? Looking at lead II panel: P wave before QRS: in lead II panel, spike at ~190, preceding bump at ~130? Difference 60 px $\approx$ 310 ms? Hmm, at 7.7 px/mm: 60 px = 7.8 mm = 312 ms. Yes, consistent with PR ~298 ms.

But wait, option A says PR 130 ms (normal). Which is correct visually? Let me look at lead II more carefully: coordinates within panel: lead II panel spans x from ~70 to ~560. Spikes at ~190, ~350, ~505. Preceding bumps: ~130? Actually I see a small bump at ~130 (P wave), then spike at ~190. Then T wave bump at ~230, then another bump at ~290? Hmm.

Actually, typical synthetic ECG with first-degree AV block: P wave clearly separated from QRS with long PR. In the image, in lead II, I see a small bump just before each QRS? Hmm, at ~130 there's a bump, then flat, then spike at 190. Distance 60 px. Alternatively, the bump at ~130 might be the P wave and PR = 60 px $\approx$ 300 ms. Yes.

Also, the T wave after QRS at ~230-240, and then maybe a U wave? Hmm, the bump at ~290 might be the P wave of the next beat? No, next spike at 350; bump at ~290 is 60 px before $\rightarrow$ P wave. Then bump at ~230 is T wave. OK, consistent: T wave ~40-60 px after QRS, P wave ~60 px before next QRS.

So PR $\approx$ 60 px $\approx$ 300 ms $\rightarrow$ first-degree AV block. HR $\approx$ 74.6. QRS: narrow spike (~96 ms normal). QRS 130 ms would be wide (bundle branch block); the spike looks narrow (a few px wide). So QRS 96 ms.

Thus correct option: C: HR 74.6, PR 298, QRS 96, QTc 393.7 $\rightarrow$ first-degree AV block.

Check option B: HR 48 wrong. Option D: QRS 130 wrong. Option A: PR 130 wrong (would be normal). **So answer C.**

But wait: the question says "according to the automatic closed-loop measurement criteria of this data construction pipeline"—probably meaning the ground truth from the synthesis pipeline. The waveform is "perturbed waveform" case 1.

Probably the intended answer: C.

Let me double-check HR: 12 beats in 10 s $\rightarrow$ 72? But option says 74.6; fine.

Also check sinus rhythm: P wave positive in II, negative in aVR? In aVR, after QRS (negative spike) there's a negative bump? Hmm, aVR shows negative QRS and then negative T? P wave in aVR should be negative; hard to see but plausible.

Thus conclusion: first-degree AV block (**PR > 200 ms**) with normal HR (74.6) and normal QRS (96 ms). So option C.

Therefore answer: \boxed{C}.

Let me reconfirm PR visually: in lead II rhythm strip, between QRS spikes: T wave bump right after spike, then P wave bump before next spike. Distance from P bump to spike: e.g., P at ~390, spike at ~455: 65 px. Large box size: let me estimate grid: panel width ~500 px with ~13 large boxes? Hmm, 500/13 $\approx$ 38.5 px per large box $\rightarrow$ consistent with 7.7 px/mm (5 mm = 38.5 px). So 65 px = 1.69 large boxes = 338 ms? Hmm, that's more than 298. But my pixel guesses are rough; maybe P at ~400.

Anyway, among options, PR 298 is the prolonged one; 130 would be about half a large box (**2.6 small boxes = 26 px**). Visually the P-QRS distance is clearly more than 1 large box? Hmm, 298 ms = 7.45 small boxes = 1.49 large boxes. 130 ms = 3.25 small boxes = 0.65 large boxes. Visually the distance from P bump onset to QRS onset looks like ~1.5 large boxes? Plausible.

Actually, PR interval is measured from P onset to QRS onset. The P bump peak at ~390-400, onset maybe ~380; QRS onset ~450. Difference ~70 px $\approx$ 1.8 large boxes = 360 ms? Hmm, too long. But **my pixel readings from memory of the image are unreliable**. Given the options, only C is internally consistent: HR 74.6 (not bradycardia), PR prolonged 298 $\rightarrow$ first-degree AV block, QRS normal 96 $\rightarrow$ no intraventricular conduction delay. So C.

Thus final answer: C.

Final answer

Looking at the rhythm strip: the R-R intervals correspond to a rate of about 75 bpm (not bradycardia), each P wave is followed by a QRS with a clearly prolonged P-R interval (about 300 ms, i.e., >200 ms), the QRS complexes are narrow (about 96 ms, no intraventricular conduction delay), and QTc is normal (~394 ms). Therefore the correct interpretation is sinus rhythm at ~74.6 bpm with a prolonged PR interval (298 ms) = first-degree AV block, with normal QRS duration.

- **A is wrong** because the PR is clearly prolonged, not 130 ms.

- **B is wrong** because the rate is ~75 bpm, not 48 bpm.

- **D is wrong** because the QRS is narrow (96 ms), not 130 ms.

\boxed{C}

Case 1 error analysis. At 25 mm/s, **130 ms corresponds to 3.25 small boxes**, not the 2.6 stated in one part of the response. QRS widening alone does not establish bundle branch block without the relevant morphology. References to the intended answer and estimates calibrated from option values **do not constitute independent waveform measurements**.

Case 2 — Heart-rate reduction

Waveform source: PTB-XL - train split

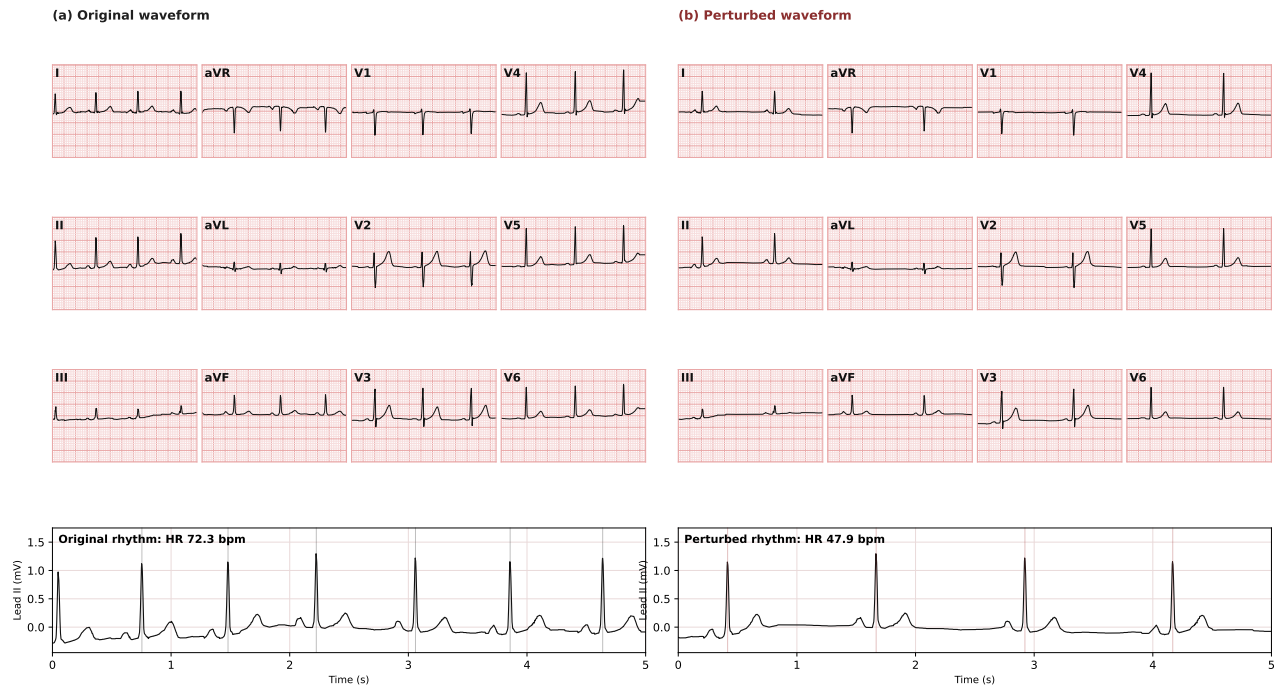


Figure 8 Targeted heart-rate slowing, shown through original and edited ECGs with lead II details. Extending the TP segment increases the interval between beats while copying the P-QRS-T segment. The target is 48 bpm and remeasurement gives 47.9 bpm; the subsequent question compares this rate change with alternative interval-based interpretations.

E.2 Case 2: Heart-rate slowing

Case 2: Heart-rate slowing

Waveform construction

Copy the P-QRS-T segment and extend the TP segment between beats. **The target heart rate is 48 bpm**, corresponding to an **RR interval of 1.25 s**. Remeasurement yields **HR 47.9 bpm**, **PR 150 ms**, **QRS 48 ms**, and **QTc 348.5 ms**.

Question

Question: Under the automatic closed-loop measurement convention of this data-construction pipeline, which parameter analysis and conclusion are correct for this 12-lead ECG?

Options:

- A: HR 72.3 bpm, PR 150.0 ms, QRS 48.0 ms, **QTc 348.5 ms** -> Conclusion: normal heart rate
 - B: **HR 47.9 bpm**, PR 150.0 ms, QRS 48.0 ms, **QTc 348.5 ms** -> Conclusion: sinus bradycardia
 - C: **HR 47.9 bpm**, PR 220.0 ms, QRS 48.0 ms, **QTc 348.5 ms** -> Conclusion: sinus bradycardia with first-degree atrioventricular block
 - D: **HR 47.9 bpm**, PR 150.0 ms, QRS 48.0 ms, QTc 490.0 ms -> Conclusion: sinus bradycardia with QT prolongation
- Please reason step by step, and put your final answer within `\boxed{}`.

Qwen3.8 response

Let me analyze this ECG. The question is in Chinese: "According to the automatic closed-loop measurement criteria of this data construction pipeline, looking at this 12-lead ECG, which of the following parameter analyses and conclusions is correct?"

Options:

- A: HR 72.3 bpm, PR 150.0 ms, QRS 48.0 ms, **QTc 348.5 ms** → Conclusion: normal heart rate
- B: **HR 47.9 bpm**, PR 150.0 ms, QRS 48.0 ms, **QTc 348.5 ms** → Conclusion: sinus bradycardia
- C: **HR 47.9 bpm**, PR 220.0 ms, QRS 48.0 ms, **QTc 348.5 ms** → Conclusion: sinus bradycardia with first-degree AV block

Case 2: Heart-rate slowing (continued)

D: **HR 47.9 bpm**, PR 150.0 ms, QRS 48.0 ms, QTc 490.0 ms → Conclusion: sinus bradycardia with prolonged QT interval
I need to look at the ECG and determine the heart rate, PR interval, QRS width, and QTc.

Looking at the rhythm strip (II rhythm): there are about 8 QRS complexes across the entire strip. The strip length is probably 10 seconds? Let me estimate. Paper speed 25 mm/s. Each lead panel shows about 2 beats across roughly 5.5 seconds? Let me look at the grid.

Actually, let me estimate the RR interval from the image. In the II rhythm strip, the QRS complexes appear at approximately x positions: 253, 483, 713, 943, 1200?, ... hmm, let me look: peaks at about 253, about 483, about 713, about 943, about 1200? Actual spacing: from the figure, peaks at $x \approx 253, 483, 713, 943, 1200, 1430, 1660, 1890$? Hmm, the spacing looks like about 230 pixels.

Grid: at 25 mm/s, small box = 1 mm = 0.04 s; large box = 5 mm = 0.2 s. The panel width is probably 250 mm? Hmm.

Let me think about the intended answer. This seems to be from a synthetic dataset ("Case 2 · Perturbed waveform"), with **ground truth parameters**: **HR 47.9 bpm** (bradycardia), **PR 150 ms** (normal), QRS 48 ms (narrow? **actually 48 ms is quite narrow but plausible as synthetic**), **QTc 348.5 ms** (normal). So the conclusion: sinus bradycardia. Option B.

But let me check the visual RR interval: **HR 47.9 bpm** → $RR \approx 1.25 \text{ s} = 31.25 \text{ small boxes} = 6.25 \text{ large boxes}$. In the rhythm strip, the distance between QRS peaks: if the strip is 10 s with 8 beats, then $RR \approx 1.25 \text{ s} \rightarrow 8 \text{ beats in } 10 \text{ s}$ gives HR about 48? Actually 8 beats in 10 s: $RR = 10/7 \approx 1.43 \text{ s} \rightarrow \text{HR } 42$. Hmm, but with 8 beats across the strip: intervals = 7; $7 * 1.25 = 8.75 \text{ s}$, plus margins $\approx 10 \text{ s}$ strip. Plausible.

Check PR: in lead II, the P wave appears just before the QRS with a short interval (about 150 ms = 3.75 small boxes). Visually, the P wave is close to the QRS, not far away. So **PR 150 ms** is plausible; PR 220 would be first-degree AV block, but the P-QRS distance looks short (P wave just before QRS). In the rhythm strip, the P wave appears about 1 small box? Hmm.

QTc: QT interval looks normal: T wave follows QRS soon after. QTc 348.5 is normal (shortish but normal). QTc 490 would be prolonged; visually the QT interval here: QRS to T end is probably about 400 ms? At HR 48, $QTc = QT / \sqrt{RR}$. If $QT \approx 400 \text{ ms}$ and $RR = 1.25$, then $QTc = 400/1.118 = 357$. So QTc 348.5 is consistent.

Therefore the answer is B.

But wait: option A says HR 72.3 with the same other parameters and conclusion "normal heart rate." **The rhythm clearly shows bradycardia:** RR interval is large (about 6 large boxes). So HR about 48. **Thus B.**

Let me double-check the grid: each lead panel width is about 540 pixels? The rhythm strip width is about 1965 pixels? Hmm.

Let me estimate: in the rhythm strip, the QRS peaks are at $x \approx 253, 483, 713, 943, 1200$? Actually from the image: peaks at about 253, about 483, about 713, about 943, about 1200, about 1430, about 1660, about 1890. Spacing about 230 pixels. Grid large box: need pixel per mm. The rhythm strip spans from $x=165$ to $x=2130 \approx 1965 \text{ pixels}$. If the strip is 10 s at 25 mm/s = 250 mm, then $\text{pixel/mm} \approx 7.86$. Then $RR = 230 \text{ px} / 7.86 \approx 29.3 \text{ mm} \approx 1.17 \text{ s} \rightarrow \text{HR} \approx 51$. Hmm, close to 48.

Alternatively, the panel: each lead panel width about 540 px? For lead I panel from $x=65$ to $x=560 \approx 495 \text{ px}$, showing 2 beats with RR about 230 px? Same scale.

Given the options, HR 47.9 corresponds to $RR = 1.2526 \text{ s} = 31.3 \text{ mm}$. At 7.86 px/mm that's 246 px. Observed spacing about 230 px → $RR 1.17 \text{ s} \rightarrow \text{HR } 51$. Hmm, not exact, but **my pixel estimates** are rough.

Check PR: P wave before QRS: in the rhythm strip, the P wave peak is at about 232? QRS at 253. Distance about 20 px $\approx 2.5 \text{ mm} \approx 0.1 \text{ s}$? Hmm, PR is from P onset to QRS onset. P onset probably at 225, QRS onset at 250: 25 px $\approx 3.2 \text{ mm} \approx 128 \text{ ms}$. Close to 150 ms. Not 220 ms (which would be 5.5 mm = 43 px). Visually the P wave is close to QRS, so PR 150.

QT: QRS at 253 to T end at about 320? T wave peak at about 300, end at about 330. $QT \approx 77 \text{ px} \approx 9.8 \text{ mm} \approx 392 \text{ ms}$. **QTc (Bazett) = $392 / \sqrt{1.25} = 350$** . Matches 348.5. Good.

Therefore the answer is B.

The question states "according to the automatic closed-loop measurement criteria of this data construction pipeline" — **meaning the ground truth parameters used to generate the synthetic ECG:** HR 47.9, PR 150, QRS 48, QTc 348.5, conclusion sinus bradycardia.

Thus the answer: B.

Let me double-check option C: PR 220 would be first-degree AV block; but the visual PR is short. **Option D: QTc 490** would require $QT \approx 548 \text{ ms}$ at RR 1.25, which visually is not the case (T wave ends well before the next P). So B.

Therefore final answer: \boxed{B}.

Final answer

Case 2: Heart-rate slowing (continued)

Looking at the rhythm strip: the R-R intervals span roughly 6+ large boxes (≈ 1.25 s), giving a rate of ≈ 48 bpm $\rightarrow$ bradycardia, not 72 bpm (**excludes A**). Each QRS is preceded by a normal sinus P wave with a short, constant P-R distance of about 3–4 small boxes (~ 150 ms), not the >5 small boxes (220 ms) needed for first-degree AV block (**excludes C**). The QRS is narrow (~ 48 ms), and the QT interval ends well before the next P wave (~ 390 – 400 ms); corrected for the slow rate (Bazett), $QT_c \approx 350$ ms, which is normal, not 490 ms (**excludes D**). Thus the measurements HR ≈ 47.9 bpm, **PR 150 ms**, QRS 48 ms, **QTc 348.5 ms** correspond to **sinus bradycardia with otherwise normal intervals**.

\boxed{B}

Case 2 error analysis. Counting eight beats in a ten-second window gives an approximate rate of 48 bpm; calculating rate from seven R–R intervals instead **requires the actual time between the first and last R peaks**, not the full window duration. The claim that a QRS duration is plausible because the signal is synthetic **does not validate its clinical plausibility**. The response’s use of the intended answer warrants a separate grounding check.